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**DOMAIN GENERALIZATION FOR
ADVERSARIAL ROBUSTNESS**

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Declaration of Authenticity

We declare that we solely conducted this specialized project, under the supervision of Dr. Nguyen An Khuong at the Faculty of Computer Science and Engineering, Vietnam National University - Ho Chi Minh City University of Technology and Mr. Van Minh Hao at University of Arkansas.

We have taken care to properly acknowledge and document all external sources and references used in the project.

If there is any instance of plagiarism, we are ready to accept the consequences. Ho Chi Minh City University of Technology - Vietnam National University HCMC will not be held responsible for any copyright violations that may have occurred during my research.

Ho Chi Minh City, December 2025

Authors

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Abstract

This thesis addresses the critical vulnerability of deep learning models to adversarial perturbations, particularly the limitation where standard defense mechanisms fail to generalize across different threat models. Our research proposes a novel perspective that treats distinct families of adversarial attacks as separate "domains," aiming to identify and quantify the "cross-attack robustness gap" inherent in current training paradigms.

We establish a comprehensive experimental framework to evaluate the transferability of robustness across diverse adversarial environments. By subjecting models trained on specific perturbation norms to a wide spectrum of unseen attack algorithms, the study systematically measures the performance degradation that occurs when shifting between these domains.

Ultimately, this quantitative analysis serves as the empirical foundation for the subsequent phase of research. By characterizing the domain shifts between attack families, we pave the way for integrating Domain Generalization principles into adversarial training. This progression aims to directly mitigate the observed robustness gaps, ensuring that future models can effectively generalize to unforeseen and heterogeneous adversarial scenarios.

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List of Abbreviations

Abbreviation	Definitions
AA	AutoAttack
AT	Adversarial Training
CW	Carlini–Wagner
DG	Domain Generalization
DNN	Deep Neural Network
FGSM	Fast Gradient Sign Method
IRM	Invariant Risk Minimization
PGD	Projected Gradient Descent
REx	Risk Extrapolation

Chapter 1

Introduction

1.1 Adversarial Robustness and the Cross-Attack Gap

1.2 Stage 1 Findings and Their Limitations

1.3 Problem Statement

1.4 Research Questions and Hypotheses

1.5 Contributions of This Work

1.6 Scope and Limitations

1.7 Report Organization

Chapter 2

Background

2.1 Preliminaries

This section introduces the basic notation and learning setting used throughout the report. Since Stage 1 already established the general background on deep neural networks, adversarial examples, adversarial attacks, and adversarial training, the present section provides a concise recap of these foundations before extending them toward the Stage 2 methodology. In particular, we focus on the concepts that are necessary for formulating adversarial training under multiple attack-induced domains.

We consider a supervised classification problem with input space $\mathcal{X} \subseteq \mathbb{R}^d$ and label space $\mathcal{Y} = \{1, \dots, K\}$. A training dataset is denoted by

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n,$$

where each sample (x_i, y_i) is drawn from an underlying data distribution P_{XY} . The goal is to learn a classifier that predicts the correct label for unseen inputs while remaining stable under adversarial perturbations. This stability requirement is central to adversarial robustness and becomes the main motivation for the multi-attack and group-aware training methods studied in later chapters.

2.1.1 Deep Neural Networks and Classification

Deep neural networks (DNNs) have become the dominant paradigm for modern classification tasks due to their ability to learn hierarchical feature representations directly from data. A DNN consists of multiple layers of nonlinear transformations, where early lay-

ers often capture low-level patterns such as edges or textures, while deeper layers encode more abstract and task-specific semantic features.

Formally, a neural network with parameters θ defines a function

$$f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^K,$$

which maps an input $x \in \mathbb{R}^d$ to a vector of K real-valued outputs. These outputs are commonly referred to as logits and represent the model's unnormalized scores for each class. The predicted class is obtained by selecting the class with the largest logit:

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} f_\theta(x)_k.$$

For probabilistic interpretation, the logits are usually passed through the softmax function

$$p_\theta(y = k \mid x) = \frac{\exp(f_\theta(x)_k)}{\sum_{j=1}^K \exp(f_\theta(x)_j)}.$$

The model is trained by minimizing the cross-entropy loss between the predicted distribution and the ground-truth label. For a single training sample (x, y) , the loss is defined as

$$\ell(f_\theta(x), y) = -\log p_\theta(y \mid x).$$

Standard empirical risk minimization trains the model by minimizing the average loss over the training set

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i).$$

Although this objective is effective for improving clean classification accuracy, it does not explicitly require the classifier to be stable under small input perturbations. In high-dimensional input spaces, a perturbation δ with small norm can still significantly change the logits and alter the predicted label:

$$\arg \max_k f_\theta(x)_k \neq \arg \max_k f_\theta(x + \delta)_k, \quad \|\delta\|_p \leq \varepsilon.$$

This phenomenon motivates the study of adversarial robustness. In Stage 1, this vulnerability was analyzed through a diverse suite of attacks, including FGSM, PGD, Carlini–Wagner, DeepFool, and AutoAttack. The results showed that robustness obtained against one attack configuration does not necessarily transfer to other attacks, revealing a cross-attack robustness gap. Stage 2 builds on this observation by treating different attack mechanisms as distinct domains and by developing group-aware training objectives to reduce this gap.

2.1.2 Adversarial examples

Adversarial examples are inputs that have been intentionally perturbed to cause a machine learning model to make incorrect predictions, while the perturbations remain small enough to preserve the original semantic label. Given a clean input $x \in \mathbb{R}^d$ with true label y , an adversarial example can be written as

$$x^{adv} = x + \delta,$$

where δ denotes the adversarial perturbation. The objective of the attacker is to find a perturbation that changes the model prediction while remaining within a constrained perturbation set. In the untargeted setting, the attacker aims to produce x^{adv} such that

$$\arg \max_k f_\theta(x^{adv})_k \neq y.$$

In contrast, a targeted attack attempts to force the model to predict a specific target class $y_{target} \neq y$:

$$\arg \max_k f_\theta(x^{adv})_k = y_{target}.$$

Most adversarial attacks constrain the perturbation by an ℓ_p -norm budget. The feasible perturbation set is commonly defined as

$$\mathcal{S}_p(\varepsilon) = \{\delta \in \mathbb{R}^d : \|\delta\|_p \leq \varepsilon\},$$

where $p \in \{2, \infty\}$ is the norm type and $\varepsilon > 0$ controls the maximum allowed perturbation size. The adversarial example is also clipped to the valid input range, typically $x^{adv} \in [0, 1]^d$ for normalized images. Therefore, the adversary searches within the fea-

sible set

$$x^{adv} \in (x + \mathcal{S}_p(\varepsilon)) \cap [0, 1]^d.$$

Under this formulation, adversarial example generation can be viewed as an inner maximization problem. For a given model f_θ , loss function ℓ , and clean sample (x, y) , the attacker seeks

$$\delta^* = \arg \max_{\delta \in \mathcal{S}_p(\varepsilon)} \ell(f_\theta(x + \delta), y).$$

The resulting adversarial example $x + \delta^*$ corresponds to an input within the allowed perturbation region that maximizes the model’s loss. In practice, this maximization is approximated by different attack algorithms, such as FGSM, PGD, Carlini–Wagner, DeepFool, or AutoAttack. Stage 1 used these attacks as a diverse evaluation suite to measure how robust a PGD-trained model remains when exposed to attacks beyond the one used during training.

A central assumption in adversarial robustness is label preservation. Although the input is perturbed, the true semantic class is assumed to remain unchanged:

$$y(x) = y(x + \delta), \quad \text{for } \|\delta\|_p \leq \varepsilon.$$

This assumption distinguishes adversarial perturbations from natural distribution shifts or semantic transformations that genuinely change the class. For example, in image classification, a small imperceptible perturbation to a CIFAR-10 image should not change a “cat” into a “dog” from a human perspective. If the model misclassifies the perturbed image, the error reflects a lack of robustness rather than ambiguity in the data.

Different attacks produce perturbations with different geometric and optimization characteristics. Gradient-based attacks such as FGSM and PGD directly use the gradient of the loss with respect to the input. Optimization-based methods such as Carlini–Wagner search for small perturbations that satisfy a misclassification objective. Boundary-based methods such as DeepFool estimate the closest decision boundary, while ensemble methods such as AutoAttack combine multiple attack strategies to provide a stronger and more reliable robustness evaluation. These differences are important because a model that performs well against one attack may still fail against another.

This observation is the key motivation for the Stage 2 formulation. Instead of treating attacks only as evaluation tools, we interpret each attack mechanism as inducing a distinct adversarial domain. For an attack e , let $\mathcal{A}_e$ denote the corresponding adversarial example generator. The adversarial sample generated by this attack can be written as

$$x_e^{adv} = \mathcal{A}_e(x, y, f_\theta).$$

The risk associated with attack domain e is then defined as

$$R_e(\theta) = \mathbb{E}_{(x,y) \sim P_{XY}} [\ell(f_\theta(x_e^{adv}), y)].$$

From this perspective, the goal is not only to minimize adversarial loss for a single attack, but also to control the variation of risks across multiple attack domains. A model trained only on one attack may achieve low $R_e(\theta)$ for the training attack but still suffer high risk under another attack e' . This mismatch gives rise to the cross-attack robustness gap, which Stage 1 empirically identified and Stage 2 aims to reduce using multi-attack and group-aware optimization methods.

2.1.3 Threat models

In this project, we consider adversarial perturbations constrained by an ℓ_p -norm budget, as introduced in the previous section. For a clean input x , the adversary searches for an adversarial example $x^{adv} = x + \delta$ such that

$$\|\delta\|_p \leq \varepsilon, \quad x^{adv} \in [0, 1]^d,$$

where $p \in \{2, \infty\}$ and ε controls the perturbation magnitude. In addition to the perturbation constraint, the adversary's access to the model determines the type of threat model. We distinguish three common settings: white-box, gray-box, and black-box attacks.

White-box threat model In the white-box setting, the adversary has full access to the model architecture, parameters, loss function, and gradients. This is the strongest and most commonly used setting for evaluating first-order robustness. Given full access to f_θ , the attacker can compute gradients of the loss with respect to the input:

$$\nabla_x \ell(f_\theta(x), y),$$

and use them to construct adversarial examples. Attacks such as FGSM and PGD are typical white-box attacks because they directly rely on input gradients to increase the classification loss.

The white-box setting is particularly important for adversarial training. During training, the model itself is used to generate adversarial examples through an inner maximization procedure. For example, PGD-based adversarial training approximates

$$\max_{\delta \in \mathcal{S}_p(\varepsilon)} \ell(f_\theta(x + \delta), y)$$

using multiple projected gradient ascent steps. Since the attack is generated using the current model’s gradients, adversarial training naturally operates in a white-box setting.

In Stage 2, the main training attacks are also considered under the white-box threat model. This allows the training procedure to generate strong adversarial samples from the current model and to optimize against them. However, as observed in Stage 1, robustness under a white-box training attack does not necessarily imply robustness against other white-box attacks with different optimization strategies, norms, or loss functions.

Gray-box threat model The gray-box setting represents an intermediate level of adversarial knowledge. In this case, the adversary may not have full access to the target model’s parameters or gradients, but may have access to related information such as the model architecture, training procedure, dataset, or a surrogate model. Adversarial examples are often generated on a surrogate model and then transferred to the target model.

Formally, let f_{θ_s} be a surrogate model and f_{θ_t} be the target model. A gray-box adversarial example can be generated as

$$x^{adv} = \mathcal{A}(x, y, f_{\theta_s}),$$

and then evaluated on the target model f_{θ_t} . The attack succeeds if

$$\arg \max_k f_{\theta_t}(x^{adv})_k \neq y.$$

This setting is important because adversarial examples often transfer across models. A perturbation generated to fool one model can also fool another model, especially when the two models share similar architectures, training data, or learned decision boundaries.

In this project, gray-box evaluation is useful for measuring transfer robustness. While white-box evaluation tests whether a model is robust to attacks generated directly against itself, gray-box evaluation tests whether robustness generalizes across source and target models. This is especially relevant to the attacks-as-domains perspective, because transfer attacks can be interpreted as another form of distribution shift: the perturbation is generated from one model’s loss landscape but evaluated on another model’s decision boundary.

Black-box threat model In the black-box setting, the adversary does not have access to the model parameters or gradients. Instead, the attacker can only query the model and observe limited outputs, such as predicted labels, confidence scores, or logits. Depending on the available output information, black-box attacks can be further divided into score-based and decision-based attacks.

In score-based attacks, the adversary can observe confidence scores or logits and use them to guide the search for adversarial examples. Square Attack is an example of a score-based black-box attack. It performs random localized perturbation updates and accepts changes that improve the attack objective, without requiring gradient information.

In decision-based attacks, the adversary only observes the final predicted label. This is an even more restrictive setting because the attacker cannot directly measure how close an input is to the decision boundary using scores or probabilities. Such attacks typically require more queries and are not the primary focus of this report.

Black-box evaluation is important because it helps detect whether a defense relies on misleading or uninformative gradients. Some defenses may appear robust against gradient-based white-box attacks but fail under gradient-free black-box attacks. This phenomenon is often associated with gradient masking, where gradients become difficult to use without the model becoming genuinely robust. Therefore, including black-box or score-based components in the evaluation suite provides a stronger robustness assessment.

2.2 Adversarial Training Deep Dive

2.2.1 Min-max formulation

2.2.2 PGD-AT and its limitations

2.2.3 Multi-attack AT approaches

2.2.4 Trade-off analysis

2.3 Distributionally Robust Optimization

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2.3.2 Group DRO

2.3.3 Exponentiated gradient on simplex

2.3.4 Regularization in overparameterized models

2.3.5 DRO in adversarial training context

2.4 Domain Generalization via Clustering

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2.5 Statistical Foundations

Paired t-test, Welch t-test

Effect size vs p-value

Why 5-seed is the minimum standard in AT

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Chapter 3

Related Work

3.1 Multi-Attack Adversarial Training

3.2 Group DRO Applied to Robustness

3.3 Domain Generalization Meets Adversarial Robustness

3.4 Cluster Discovery in Robust Training Pipelines

3.5 The Remaining Gap and Motivation for This Work

Chapter 4

Proposed Methodology

4.1 Methodology Overview

Stage 1 showed that adversarial robustness does not transfer uniformly across attack families. A model trained against one threat model may remain vulnerable to attacks with different perturbation geometries, optimization procedures, or loss objectives. This motivates the Stage 2 methodology: instead of viewing all adversarial examples as samples from a single homogeneous distribution, we treat attack-induced examples as coming from multiple adversarial domains and design training objectives that can emphasize weak domains or latent hard groups while preserving the stability of the uniform multi-attack baseline.

This chapter develops the proposed framework in five steps. First, Section 4.2 formulates multi-attack adversarial training under the attacks-as-domains perspective. Second, Section 4.3 defines Uniform Multi-Attack ERM, which serves as one of the main baseline beside single attack adversarial training like PGD or DDN. Third, Section 4.4 introduces AttackDRO, a GroupDRO formulation over attack identities. Fourth, Section 4.5 extends this to AttackDRO++, which discovers latent adversarial groups through clustering. Section 4.6 introduces gradient fingerprints to enrich group discovery with optimization-level information, and Section 4.7 introduces the uniform-anchored Cluster-DRO objective to stabilize training.

The final method evaluated in the main experiments is **AttackDRO++ with Gradient Fingerprints and Uniform Anchor**. In the experimental notation this corresponds to `attackdro_pp_anchor035_gradfp_online_k4`, where `anchor035` indicates that the Cluster-DRO loss receives weight $\lambda_{\text{DRO}} = 0.35$ and the uniform multi-attack ERM loss serves as the stabilizing anchor.

4.2 Multi-Attack Training as a Domain Problem

4.2.1 Setup and Notation

Let

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n \quad (4.1)$$

denote the supervised training dataset, where $\mathbf{x}_i \in \mathcal{X}$ is an input image and $y_i \in \{1, \dots, C\}$ is its class label. Let f_θ be a classifier parameterized by θ , and let $\ell(f_\theta(\mathbf{x}), y)$ denote the cross-entropy loss.

We denote the logits produced by the model as

$$\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^C,$$

and the penultimate-layer representation as

$$\mathbf{h} = \phi_\theta(\mathbf{x}) \in \mathbb{R}^{d_h}.$$

The representation $\phi_\theta(\cdot)$ is later used for cluster discovery.

We define a set of source attack domains

$$\mathcal{A}_{\text{src}} = \{A_1, A_2, \dots, A_M\}, \quad (4.2)$$

where each attack operator A_m is characterized by a threat model, perturbation budget, optimization procedure, number of inner steps, and loss objective. Given a clean sample $(\mathbf{x}_i, y_i)$, the adversarial example produced by source attack A_m is

$$\mathbf{x}_i^{(m)} = A_m(f_\theta, \mathbf{x}_i, y_i). \quad (4.3)$$

In the main experiments, we use two source attacks: a PGD- ℓ_∞ attack and a DDN- ℓ_2 attack. This choice exposes the model to two common adversarial norm families during training. Additional attacks are reserved as heldout evaluation attacks and are never used for optimization.

4.2.2 Multi-Attack Risk and Its Limitations

The empirical adversarial risk under source attack A_m is

$$R_m(\theta) = \frac{1}{n} \sum_{i=1}^n \ell\left(f_\theta\left(\mathbf{x}_i^{(m)}\right), y_i\right). \quad (4.4)$$

A standard multi-attack objective minimizes the average risk across source attacks:

$$\min_{\theta} \frac{1}{M} \sum_{m=1}^M R_m(\theta). \quad (4.5)$$

This objective improves attack diversity compared with single-attack adversarial training. However, it treats all source attacks and all adversarial examples with equal importance. This creates two limitations.

First, if one source attack consistently induces higher loss than another, equal weighting may underemphasize the harder attack. Second, adversarial examples within the same attack are not homogeneous. Two samples generated by the same algorithm can differ substantially in classification margin, representation, and optimization direction. These within-attack differences are not captured by Equation (4.5).

The central goal of this chapter is therefore to construct adaptive objectives that can identify and upweight hard groups while preserving the stability of the uniform multi-attack training signal.

4.3 Uniform Multi-Attack ERM

Uniform Multi-Attack ERM serves as the main baseline for this work. It trains on multiple source attacks within the same minibatch but assigns equal weight to all generated adversarial examples. This makes it a strong and simple reference point for evaluating whether adaptive reweighting provides additional benefit.

Given a minibatch

$$\mathcal{B} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N,$$

we partition it into M disjoint subsets

$$\mathcal{B}_1, \dots, \mathcal{B}_M, \quad \mathcal{B}_m \cap \mathcal{B}_{m'} = \emptyset \text{ for } m \neq m', \quad \bigcup_{m=1}^M \mathcal{B}_m = \mathcal{B}.$$

Each subset $\mathcal{B}_m$ is assigned to one source attack A_m . For each $(\mathbf{x}_i, y_i) \in \mathcal{B}_m$, we generate

$$\mathbf{x}_i^{\text{adv}} = A_m(f_\theta, \mathbf{x}_i, y_i).$$

The adversarial examples from all source attacks are then concatenated into one mixed adversarial batch. The uniform multi-attack loss is

$$\mathcal{L}_{\text{ERM}} = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(\mathbf{x}_i^{\text{adv}}), y_i). \quad (4.6)$$

In our main setting, the source attacks are PGD- ℓ_∞ and DDN- ℓ_2 , so the model is exposed to both ℓ_∞ and ℓ_2 perturbation geometries during training. However, the objective in Equation (4.6) treats all adversarial examples equally. It does not explicitly identify which attack, class, or latent subgroup is harder. This limitation motivates the adaptive reweighting methods introduced in the following sections.

4.4 AttackDRO: Group DRO Over Attack Identities

AttackDRO extends Uniform Multi-Attack ERM by applying Group Distributionally Robust Optimization (GroupDRO) over attack identities. Each source attack A_m is treated as one group. The method maintains an attack-group weight vector

$$\mathbf{q} = (q_1, \dots, q_M) \in \Delta^M, \quad (4.7)$$

where Δ^M is the M -dimensional probability simplex.

For each attack group m , the group loss is

$$\mathcal{L}_m = \frac{1}{|\mathcal{B}_m|} \sum_{i \in \mathcal{B}_m} \ell(f_\theta(\mathbf{x}_i^{\text{adv}}), y_i). \quad (4.8)$$

The AttackDRO objective is then

$$\mathcal{L}_{\text{AttackDRO}} = \sum_{m=1}^M q_m \mathcal{L}_m. \quad (4.9)$$

The group weights are updated using an exponentiated-gradient (EG) rule. Let s_m be the difficulty score of attack group m . The update is

$$q_m \leftarrow q_m \exp(\eta_q s_m), \quad m = 1, \dots, M, \quad (4.10)$$

followed by normalization and floor projection:

$$q_m \geq q_{\min}, \quad \sum_{m=1}^M q_m = 1. \quad (4.11)$$

The floor constraint prevents any attack group from being completely ignored.

In our formulation, the difficulty score is based on the classification margin. For a sample i with logits $\mathbf{z}_i$, define the margin as

$$r_i = z_{i,y_i} - \max_{c \neq y_i} z_{i,c}. \quad (4.12)$$

A large positive margin indicates a confident correct prediction, while a small or negative margin indicates a hard or misclassified sample. We convert this margin into a per-sample difficulty score using

$$s_i = \text{softplus}(-r_i/\tau) = \log(1 + \exp(-r_i/\tau)), \quad (4.13)$$

where $\tau > 0$ is a temperature parameter. The group-level difficulty score is the average difficulty within group m :

$$s_m = \frac{1}{|\mathcal{B}_m|} \sum_{i \in \mathcal{B}_m} s_i. \quad (4.14)$$

AttackDRO can adaptively emphasize whichever source attack is currently harder. However, it relies on a coarse grouping assumption: all examples generated by the same attack are treated as one homogeneous group. In practice, two PGD- ℓ_∞ examples may have very different margins, representations, and gradient directions. Conversely, examples generated by different attacks may share similar failure modes. Thus, the EG update can only adapt at the attack level and may miss finer-grained difficulty structure within and across attacks. The cluster-based extension in the next section addresses this limitation.

4.5 AttackDRO++: Group DRO Over Discovered Clusters

AttackDRO++ replaces the manually defined attack groups in AttackDRO with latent clusters discovered from adversarial examples. The central hypothesis is that attack identity is too coarse as a grouping criterion. Two adversarial examples generated by different attacks may share the same failure mode, while two examples generated by the same attack may have very different margins, representations, and optimization direc-

tions. Cluster-based grouping allows the DRO objective to adapt to hard subgroups that may cut across attack boundaries.

For each adversarial example $\mathbf{x}_i^{\text{adv}}$, the model extracts a representation vector

$$\mathbf{h}_i = \phi_\theta(\mathbf{x}_i^{\text{adv}}), \quad (4.15)$$

and a clustering model assigns the example to a latent group

$$c_i \in \{1, \dots, K\}. \quad (4.16)$$

Let $\mathcal{B}_k$ denote the subset of the current minibatch whose adversarial examples are assigned to cluster k :

$$\mathcal{B}_k = \{i \in \{1, \dots, N\} : c_i = k\}. \quad (4.17)$$

The cluster loss is

$$\mathcal{L}_k = \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} \ell(f_\theta(\mathbf{x}_i^{\text{adv}}), y_i). \quad (4.18)$$

AttackDRO++ maintains a cluster-level weight vector

$$\mathbf{q} = (q_1, \dots, q_K) \in \Delta^K. \quad (4.19)$$

Because not every cluster appears in every minibatch, the cluster weights are renormalized over the clusters present in the current minibatch. Let

$$\mathcal{C}_{\mathcal{B}} = \{k \in \{1, \dots, K\} : |\mathcal{B}_k| > 0\} \quad (4.20)$$

denote the set of present clusters. For each $k \in \mathcal{C}_{\mathcal{B}}$, we define

$$\tilde{q}_k = \frac{q_k}{\sum_{j \in \mathcal{C}_{\mathcal{B}}} q_j}. \quad (4.21)$$

The cluster-DRO loss is then

$$\mathcal{L}_{\text{ClusterDRO}} = \sum_{k \in \mathcal{C}_{\mathcal{B}}} \tilde{q}_k \mathcal{L}_k. \quad (4.22)$$

The cluster weights are updated using the same exponentiated-gradient principle as in AttackDRO, but now at the cluster level rather than the attack level. Given a cluster-

level difficulty score

$$s_k = \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} s_i, \quad k \in \mathcal{C}_{\mathcal{B}}, \quad (4.23)$$

the update increases the weight of clusters with higher average difficulty. This allows AttackDRO++ to emphasize hard latent groups even when they do not align exactly with the original attack identities.

Since model representations evolve during training, cluster assignments must also be updated. AttackDRO++ therefore maintains a feature bank $\mathcal{F}$, which stores detached augmented feature vectors from recent minibatches. When the refresh condition is met, K-means is refitted on $\mathcal{F}$, and subsequent minibatches are assigned using the updated clusterer. Before the first K-means refresh, cluster assignments are bootstrapped randomly.

The refresh condition is treated as a policy rather than a fixed part of the method. It can be defined in terms of epochs or training steps. In the main experiments, we use an epoch-based refresh schedule with $R = 2$ epochs. After each refresh, the feature bank is reset or updated according to the memory policy, and the cluster weights are refreshed according to the chosen q -refresh policy.

4.6 Augmented Clustering with Gradient Fingerprints

Representation-space clustering alone may not fully capture adversarial difficulty. Two adversarial examples can be close in feature space while inducing different optimization behavior. To capture this additional signal, AttackDRO++ constructs an augmented clustering feature that combines representation similarity, class information, and gradient-level training dynamics.

For an adversarial example x_i^{adv} , let

$$(z_i, h_i) = f_{\theta}(x_i^{\text{adv}})$$

denote the logits and penultimate-layer feature, and let

$$p_i = \text{softmax}(z_i)$$

be the predicted class distribution.

The first component is the normalized representation feature,

$$\bar{h}_i = \text{norm}(h_i),$$

which captures semantic similarity in the model feature space.

The second component is a class-aware label feature,

$$\lambda_{\text{lbl}} \text{onehot}(y_i),$$

which provides weak label structure during clustering.

The third component is the gradient fingerprint. For a linear classifier head

$$f_\theta(\cdot) = W\phi_\theta(\cdot) + b, \quad W \in \mathbb{R}^{C \times d_h},$$

the per-example gradient of the cross-entropy loss with respect to the classifier weight matrix is

$$\nabla_W \ell(f_\theta(x_i^{\text{adv}}), y_i) = (p_i - \text{onehot}(y_i)) \otimes h_i \in \mathbb{R}^{C \times d_h}.$$

We use this quantity as a gradient proxy and denote it by

$$G_i = (p_i - \text{onehot}(y_i)) \otimes h_i.$$

Intuitively, G_i describes the training direction induced by sample i . Thus, two adversarial examples may be grouped together not only because they are close in representation space, but also because they induce similar optimization updates.

The raw gradient proxy has dimension Cd_h . For CIFAR-10 with a ResNet-18 backbone, this corresponds to $10 \times 512 = 5120$ dimensions. To make the fingerprint more compact and efficient for clustering, we use a fixed random projection matrix

$$P \in \mathbb{R}^{Cd_h \times d_{\text{proj}}},$$

and define the projected gradient fingerprint as

$$g_i = \text{vec}(G_i)P \in \mathbb{R}^{d_{\text{proj}}}.$$

In our default setting, $d_{\text{proj}} = 128$. The projection is fixed throughout training and introduces no additional learnable parameters.

The final augmented clustering feature is

$$\psi_i = [\text{norm}(h_i) \parallel \lambda_{\text{lbl}} \text{onehot}(y_i) \parallel \lambda_{\text{grad}} \text{norm}(g_i)] .$$

The three components play complementary roles. The representation feature captures semantic similarity, the label feature provides weak class-aware structure, and the gradient fingerprint captures optimization-level similarity. This allows AttackDRO++ to discover latent adversarial groups that reflect both where examples lie in feature space and how they influence training.

4.7 Uniform-Anchored Training Objective

A pure Cluster-DRO objective can become unstable when the cluster weights $\mathbf{q}$ concentrate too strongly on a small number of hard clusters. We refer to this failure mode as *q-collapse*. In the extreme case, the objective approaches a worst-cluster loss, where most of the training signal is dominated by one cluster. This behavior is undesirable in multi-attack adversarial training because the hardest cluster may change after each cluster refresh, and over-emphasizing one cluster can reduce robustness on the remaining attacks or groups.

To stabilize training, AttackDRO++ uses a uniform-anchored Cluster-DRO objective that interpolates between the uniform multi-attack ERM loss and the adaptive cluster-DRO loss:

$$\mathcal{L}_{\text{AttackDRO++}} = (1 - \lambda_{\text{DRO}}) \mathcal{L}_{\text{ERM}} + \lambda_{\text{DRO}} \mathcal{L}_{\text{ClusterDRO}} . \quad (4.24)$$

Here, $\lambda_{\text{DRO}} \in [0, 1]$ controls the strength of the Cluster-DRO correction. When $\lambda_{\text{DRO}} = 0$, the method reduces to uniform multi-attack ERM. When $\lambda_{\text{DRO}} = 1$, the method becomes pure Cluster-DRO.

Equivalently, Equation (4.24) can be written as

$$\mathcal{L}_{\text{AttackDRO++}} = \mathcal{L}_{\text{ERM}} + \lambda_{\text{DRO}} (\mathcal{L}_{\text{ClusterDRO}} - \mathcal{L}_{\text{ERM}}) . \quad (4.25)$$

This form shows that λ_{DRO} scales the difference between the cluster-reweighted loss and the uniform loss. If the cluster-DRO loss is close to the uniform loss, the correction is small. If some clusters are substantially harder, the correction becomes larger, but its influence remains controlled by λ_{DRO} .

In the main experiments, we use $\lambda_{\text{DRO}} = 0.35$, referred to as the Anchor35 setting. This value keeps the uniform multi-attack objective as the dominant stabilizing term while

still allowing the model to adapt toward harder clusters. Intuitively, this suggests an inverted-U behavior: a small amount of Cluster-DRO pressure can improve robustness by emphasizing hard groups, whereas excessive pressure may overfit unstable clusters and harm the balance across attacks.

A second stabilization mechanism is the floor projection applied to the cluster weights. After each exponentiated-gradient update, $\mathbf{q}$ is normalized and projected so that every cluster keeps at least $q_{\min}$ mass:

$$q_k \geq q_{\min}, \quad \sum_{k=1}^K q_k = 1. \quad (4.26)$$

This prevents any cluster from being completely ignored. In our default setting, $q_{\min} = 0.03$. Together, the uniform anchor and the floor projection ensure that training receives both a stable per-sample ERM signal and an adaptive cluster-level correction.

Finally, the q -update is delayed during the early training phase. For the first T_w epochs, $\mathbf{q}$ remains uniform while the model representation and initial cluster assignments become more reliable. After this warmup period, the cluster weights are updated using the exponentiated-gradient rule. In the main configuration, $T_w = 3$ epochs.

4.8 Complete Training Framework

Algorithm 1 summarizes the complete AttackDRO++ training pipeline. The method combines mixed adversarial batch generation, augmented feature construction, cluster-level reweighting, q -updates, and a uniform-anchored loss. Table 4.1 lists the default hyperparameters used in the main experiments.

Table 4.1: Default hyperparameters for AttackDRO++ in the main experiments.

Hyperparameter	Symbol	Value	Role
DRO anchor strength	λ_{DRO}	0.35	Interpolates between uniform ERM and Cluster-DRO
Number of clusters	K	4	Controls latent group granularity
q learning rate	η_q	3×10^{-4}	Step size for exponentiated-gradient update
q -floor	$q_{\min}$	0.03	Minimum mass retained by each cluster
q -warmup	T_w	3 epochs	Delays adaptive reweighting until clusters are more reliable
Gradient projection dim.	d_{proj}	128	Dimension of projected gradient fingerprint
Gradient weight	λ_{grad}	0.1	Weight of GradFP component in ψ
Label weight	λ_{lbl}	0.1	Weight of label component in ψ
Feature bank size	$ \mathcal{F} $	4,096	Buffer used to refit K-means
Cluster refresh interval	R	2 epochs	Default K-means refit frequency
q -refresh policy	–	reset	Resets $\mathbf{q}$ to uniform after cluster refresh
Bootstrap assignment	–	random	Used before the first K-means refit

Algorithm 1: AttackDRO++ with Gradient Fingerprints and Uniform Anchor

Input: Training set $\mathcal{D}_{\text{train}}$, model f_θ , source attacks $\mathcal{A}_{\text{src}} = \{A_1, \dots, A_M\}$, Cluster count K , anchor weight λ_{DRO} , learning rates η_θ, η_q , q -floor $q_{\min}$, Label weight λ_{lbl} , gradient weight λ_{grad} , projection dimension d_{proj} , Refresh interval R , warmup epochs T_w

Output: Trained model f_θ^*

```
1 Initialize  $\mathbf{q} \leftarrow (1/K, \dots, 1/K)$ 
2 Initialize feature bank  $\mathcal{F} \leftarrow \emptyset$ 
3 Initialize K-means clusterer as unavailable
4 Sample fixed projection matrix  $P \in \mathbb{R}^{C d_h \times d_{\text{proj}}}$ 
5 for epoch  $e = 1, \dots, E$  do
6   for each minibatch  $(x, y)$  of size  $N$  do
7     Generate mixed adversarial batch  $x^{\text{adv}}$  using  $\mathcal{A}_{\text{src}}$ 
8     Compute logits and features  $(z, h) \leftarrow f_\theta(x^{\text{adv}})$ 
9     Build augmented features  $\psi \leftarrow \text{BuildFeature}(z, h, y, P)$ 
10    Assign cluster ids  $c \leftarrow \text{AssignCluster}(\psi, \text{Clusterer}, K)$ 
11    Compute uniform loss  $\mathcal{L}_{\text{ERM}}$ 
12    Compute cluster-DRO loss  $\mathcal{L}_{\text{ClusterDRO}}$ 
13    Compute final loss
        
$$\mathcal{L} = (1 - \lambda_{\text{DRO}})\mathcal{L}_{\text{ERM}} + \lambda_{\text{DRO}}\mathcal{L}_{\text{ClusterDRO}}.$$

14    Update model parameters  $\theta$  using SGD
15    if  $e > T_w$  then
16      | Update  $\mathbf{q}$  using exponentiated gradient and floor projection
17    end
18    Add detached features  $\psi$  to feature bank  $\mathcal{F}$ 
19  end
20  if refresh condition is met then
21    | Refresh K-means clusterer using  $\text{KMeans}(\mathcal{F}, K)$ 
22    | Apply  $q$ -refresh policy to  $\mathbf{q}$  // default: reset to uniform
23    | Reset or update feature bank  $\mathcal{F}$  according to the memory policy
24  end
25 end
26 return  $f_\theta^*$ 
```

The refresh condition is written as a policy rather than a fixed operation. It can be instantiated using either epoch-based or step-based schedules. Unless otherwise specified, the

main experiments use an epoch-based schedule with $R = 2$ epochs and reset $\mathbf{q}$ to the uniform distribution after each cluster refresh.

4.8.1 Progression from Stage 1 to the Final Method

This methodology directly extends the attacks-as-domains framework from Stage 1, which diagnosed the cross-attack robustness gap. Stage 2 moves from diagnosis to intervention. The progression is:

1. **Single-AT**: trains against one source attack and can leave a large cross-attack robustness gap.
2. **Uniform ERM**: trains on multiple source attacks with equal weighting; it is strong but cannot adapt to weak groups.
3. **AttackDRO**: reweights attack-level groups, but remains limited by the assumption that each attack group is homogeneous.
4. **AttackDRO++**: discovers latent cross-attack groups using K-means over adversarial representations.
5. **Gradient Fingerprints**: enrich clustering with optimization-level difficulty information.
6. **Uniform Anchor**: stabilizes Cluster-DRO by preserving the strong uniform multi-attack training signal.

The evaluation discipline from Stage 1 is preserved throughout: source attacks are fixed during training, heldout attacks are never used for optimization, and results are reported both per attack and through aggregate robustness metrics.

4.8.2 Implementation and Computational Considerations

4.9 Chapter Summary

This chapter introduced the proposed methodology for improving cross-attack adversarial robustness. Starting from the attacks-as-domains formulation, we defined Uniform Multi-Attack ERM as the main baseline, then introduced AttackDRO (GroupDRO over attack identities) and AttackDRO++ (GroupDRO over latent clusters). The gradient fingerprint augments clustering with optimization-level difficulty information, enabling discovery of hard subgroups that cut across attack boundaries. The uniform-anchored objective balances the adaptivity of Cluster-DRO with the stability of the uniform baseline.

The main proposed method is **AttackDRO++ with Gradient Fingerprints and Uniform Anchor**. It is motivated by the hypothesis that gradient-fingerprint clusters dis-

cover adversarial difficulty structure that generalizes beyond the two source attacks to held-out attack families. Chapter 6 evaluates this hypothesis across 20 random seeds.

Chapter 5

Experimental Setup

5.1 Dataset and Preprocessing

The experiments are conducted on CIFAR-10, a standard image-classification benchmark consisting of ten object categories. Each image has spatial resolution (32×32) with three color channels. The dataset is divided into training, validation, and test subsets. In our experimental pipeline, 40,000 images are used for training, 10,000 images are used for validation and checkpoint selection, and 10,000 images are reserved for final test-set evaluation.

The validation set is used only for model selection and hyperparameter monitoring. Final reported numbers are computed on the test set after the checkpoint has been selected. This separation is important because adversarial robustness estimates can vary across random seeds and training checkpoints; using a held-out validation set reduces the risk of selecting a checkpoint based on test-set performance. All images are normalized using the same preprocessing pipeline across methods. During training, standard CIFAR-10 data augmentation is applied to the clean images before adversarial examples are generated. The augmentation includes random cropping with padding and random horizontal flipping. During validation and testing, no stochastic augmentation is used. This ensures that differences between methods are caused by the training objective rather than by differences in preprocessing or evaluation.

The same dataset split, preprocessing pipeline, and evaluation dataloaders are used for all compared methods. This includes single-attack adversarial training baselines, Multi-Attack training, AttackDRO, AttackDRO++, and the proposed AttackDRO++ variant with gradient fingerprints and uniform anchoring.

5.2 Model Architecture

The primary architecture used in this study is ResNet-18, based on the residual learning framework of He et al. [1], adapted for CIFAR-10 resolution. ResNet-18 provides a useful balance between expressiveness and computational cost: it is large enough to support nontrivial adversarial training experiments, but still efficient enough to run repeated-seed comparisons and ablation studies.

The classifier is denoted by (f_θ) . For an input image (x) , the model outputs logits

$$z = f_\theta(x) \in \mathbb{R}^C,$$

where $(C=10)$ is the number of CIFAR-10 classes. In addition to logits, the proposed method requires access to the penultimate-layer representation

$$h = \phi_\theta(x) \in \mathbb{R}^{d_h}.$$

This feature vector is used for clustering adversarial examples in AttackDRO++ and for constructing the gradient-fingerprint augmented feature described in Chapter 4. For ResNet-18 on CIFAR-10, the penultimate representation dimension is $(d_h = 512)$. Therefore, the raw classifier-head gradient proxy has dimension $(C \cdot d_h = 10 \times 512 = 5120)$, which is later reduced by random projection.

Unless otherwise stated, all main experiments use the same ResNet-18 architecture and differ only in the training objective. This controlled setup ensures that performance differences can be attributed to the training method rather than to architectural changes. Higher-capacity architectures are treated as scaling checks or future extensions rather than the main basis for comparison.

5.3 Attack Suite and Configuration

Stage 2 evaluates whether robustness learned from a small set of source attacks generalizes to a broader attack suite. The attack suite is divided into two groups: source attacks used during training and evaluation attacks used only for validation or final testing.

5.3.1 Training attacks

The main training setup uses two source attacks:

$$\mathcal{A}_{\text{src}} = \{\text{PGD-}\ell_\infty, \text{DDN-}\ell_2\}.$$

PGD- ℓ_∞ represents the standard first-order ℓ_∞ adversarial training setting. It uses cross-entropy loss and perturbation budget $\varepsilon_\infty = 8/255$. DDN- ℓ_2 is based on the Decoupled Direction and Norm attack of Rony et al. [2]. DDN is an efficient gradient-based ℓ_2 attack that decouples the perturbation direction from its norm: the direction is updated using gradient information, while the perturbation norm is adapted depending on whether the current example is already adversarial.

Table 5.1 summarizes the source attacks used during training. Both attacks are used to generate mixed adversarial minibatches for Multi-ATtack ERM, AttackDRO, AttackDRO++, and the proposed AttackDRO++ Anchor35 GradFP method.

Table 5.1: Training attack configuration.

Attack	Norm	Budget	Steps	Loss / objective	Role
PGD- ℓ_∞	ℓ_∞	8/255	20	Cross-entropy	Source attack
DDN- ℓ_2	ℓ_2	0.5	20	DDN objective	Source attack

For Multi-ATtack training, each minibatch is split across the source attacks and adversarial examples are generated using the assigned attack. The same mixed-batch source attack construction is also used by AttackDRO, AttackDRO++, and the proposed AttackDRO++ Anchor35 GradFP method. Thus, the compared methods share the same adversarial data generation process and differ mainly in how they weight or group the generated examples.

5.3.2 Evaluation Attacks

The evaluation suite includes both source attacks and held-out attacks. The two source attacks used during training are also included in validation and test evaluation, while the remaining attacks are held out from training. This protocol directly tests the attacks-as-domains hypothesis: a robust model should not only perform well on the attacks used for optimization, but should also generalize to attacks with different perturbation geometries and optimization dynamics.

For validation and final testing, we evaluate each checkpoint using the attack suite in Table 5.2. The main aggregate metric, denoted Mean(8), is computed over the eight attacks in the table. AutoAttack is reported separately and is therefore not included in Mean(8).

Table 5.2: Validation and test attacks used for Mean(8).

Attack	Norm	Budget	Steps	Group
FGSM-RS	ℓ_∞	8/255	1	Held-out ℓ_∞
PGD- ℓ_∞	ℓ_∞	8/255	20	Source
TPGD	ℓ_∞	8/255	10	Held-out ℓ_∞
MI-FGSM	ℓ_∞	8/255	10	Held-out ℓ_∞
PGD- ℓ_2	ℓ_2	0.5	10	Held-out ℓ_2
DDN- ℓ_2	ℓ_2	0.5	20	Source
DeepFool- ℓ_2	ℓ_2	0.5	50	Held-out ℓ_2
CW- ℓ_2	ℓ_2	0.5	200	Held-out ℓ_2

For aggregate analysis, we report four summary metrics derived from the same eight attacks: Seen, Held-out ℓ_∞ , Held-out ℓ_2 , and Mean(8). Their exact definitions are given in Section 5.7.

AutoAttack configuration AutoAttack is used as an additional standardized ℓ_∞ stress test following Croce and Hein [3]. It is not included in Mean(8), because it is computationally more expensive and follows a separate evaluation protocol.

The AutoAttack configuration uses the standard ℓ_∞ setting with perturbation budget $\varepsilon_\infty = 8/255$. In early experiments and ablations, AutoAttack is sometimes evaluated on a subset of 512 test samples as a sanity check. These 512-sample results are useful for detecting severe robustness collapse, but they are not used as the primary ranking criterion because the sample size is smaller and the variance is higher. For the final stress test, AutoAttack is evaluated on the full test set for selected representative seeds.

Table 5.3: AutoAttack configuration.

Attack	Norm	Budget	Role
AutoAttack	ℓ_∞	8/255	Separate stress test

5.4 Training Configuration

All methods are trained under a controlled adversarial training pipeline. Unless otherwise stated, the optimizer, model architecture, dataset split, preprocessing pipeline, and

evaluation protocol are kept fixed across methods. This ensures that performance differences mainly come from the training objective rather than from implementation or evaluation differences.

The primary training configuration is summarized in Table 5.4. All main experiments use ResNet-18 on CIFAR-10 with stochastic gradient descent (SGD), momentum, weight decay, mixed-precision training, and a multi-step learning-rate schedule. The main methods are trained for 50 epochs. During training, each minibatch contains a balanced mixture of source attacks. In the two-source setting, approximately half of the minibatch is assigned to PGD- ℓ_∞ and the other half to DDN- ℓ_2 .

Table 5.4: General training configuration used for the main experiments.

Component	Setting	Description
Dataset	CIFAR-10	40k train / 10k validation / 10k test.
Backbone	ResNet-18	CIFAR-10 adapted residual network.
Training epochs	50	Main training budget.
Batch size	128	Used for all main methods.
Optimizer	SGD	Momentum-based stochastic gradient descent.
Momentum	0.9	Standard SGD momentum.
Weight decay	5×10^{-4}	Weight regularization.
Initial learning rate	0.1	Decayed during training.
Learning-rate schedule	Multi-step	Same schedule for all compared methods.
Mixed precision	Enabled	Reduces memory usage and training time.
Gradient clipping	Enabled	Stabilizes adversarial training.
Checkpoint selection	Validation robust average	Test set is used only after selection.

Checkpoint selection is based on validation robustness rather than test performance. After training, the selected checkpoint is evaluated once on the test set. This protocol is applied consistently to the uniform baseline and all AttackDRO variants.

Table 5.5 summarizes the main training methods. All multi-attack methods use the same source attacks and mixed-batch adversarial example generation; they differ only in how adversarial examples are weighted or grouped during optimization.

Table 5.5: Summary of compared training methods.

Method	Grouping / weighting strategy	Experimental role
PGD-AT	One source attack only	Specialist ℓ_∞ baseline.
DDN-AT	One source attack only	Specialist ℓ_2 baseline.
Multi-AT	Uniform sample weighting over mixed PGD- ℓ_∞ and DDN- ℓ_2 batches	Main strong baseline.
AttackDRO	GroupDRO over attack identity	Coarse attack-level DRO baseline.
AttackDRO++	GroupDRO over discovered clusters	Cluster-based group discovery baseline.
AttackDRO++ (Ours)	Uniform-anchored Cluster-DRO with gradient fingerprints	Proposed method.

For the proposed method, cluster assignments are refreshed periodically using a feature bank. The default refresh interval is two epochs. The cluster weight vector $\mathbf{q}$ is updated only after a warmup period, so that early noisy representations do not dominate adaptive reweighting.

5.5 Hyperparameter Choices and Ablation Factors

Table 5.6: Default hyperparameters for AttackDRO++ (Ours).

Hyperparameter	Symbol	Default value	Purpose
Source attacks	$\mathcal{A}_{\text{src}}$	PGD-20 _{CE} , DDN- ℓ_2	Training attack domains.
Number of clusters	K	4	Latent group granularity.
DRO anchor strength	λ_{DRO}	0.35	Balances ERM and Cluster-DRO.
q learning rate	η_q	3×10^{-4}	Step size for q update.
q floor	q_{min}	0.03	Prevents ignored clusters.
q warmup	T_w	3 epochs	Delays early reweighting.
Cluster refresh interval	R	2 epochs	K-means refresh frequency.
Feature bank size	$ \mathcal{F} $	4096	Stores detached features.
Gradient projection dim.	d_{proj}	128	Compresses GradFP features.
Gradient feature weight	λ_{grad}	0.1	Weights GradFP features.
Label feature weight	λ_{lbl}	0.1	Weights label features.
Clusterer	—	K-means	Assigns latent groups.
q refresh policy	—	Reset after refresh	Avoids stale cluster weights.

The number of clusters is set to $K = 4$ as a moderate grouping resolution. Too few clusters may fail to capture meaningful difficulty structure, while too many clusters may fragment the minibatch and produce noisy group-loss estimates. The default anchor strength $\lambda_{\text{DRO}} = 0.35$ is chosen to keep the Multi-ATtack loss as the dominant stabilizing signal while still allowing cluster-level adaptive correction.

The q -floor and warmup settings are included for stability. The floor projection prevents any cluster from being completely ignored, while the warmup period prevents the exponentiated-gradient update from acting on unreliable early cluster assignments. The feature bank and refresh interval control how quickly the clusterer adapts to evolving model representations. In the default setting, refreshing every two epochs provides a balance between stability and adaptivity.

The gradient fingerprint is projected to $d_{\text{proj}} = 128$ dimensions for computational efficiency. For CIFAR-10 with ResNet-18, the classifier-head gradient proxy has raw dimension $Cd_h = 10 \times 512 = 5120$. The fixed random projection reduces this dimension while retaining useful information about the optimization direction induced by each adversarial example.

Table 5.7 lists the main hyperparameters varied in ablation experiments. These ablations are used in Chapter 6 to justify the default configuration and to separate the effect of the proposed components from the strong uniform multi-attack baseline.

Table 5.7: Hyperparameters varied in ablation studies.

Ablation factor	Values / variants	Purpose
Anchor strength	$\lambda_{\text{DRO}} \in \{0.20, 0.35, 0.50\}$	Tests the uniform-anchor trade-off.
Cluster count	$K \in \{2, 4, 6, 10\}$	Tests sensitivity to group granularity.
q update	update q vs. uniform q	Tests adaptive reweighting.
Refresh schedule	1 epoch vs. 2 epochs	Tests cluster-refresh frequency.
Gradient fingerprint	With vs. without GradFP	Tests optimization-level features.
Random clusters	Learned vs. random assignment	Negative control for clustering.

5.6 Statistical Significance Protocol

To ensure that the reported improvements are not caused by seed-specific randomness, this study applies a repeated-seed statistical protocol. Each method is trained and evaluated under the same set of seeds, dataset split, architecture, attack suite, and evaluation configuration. The seed-level result is used as the basic unit of analysis.

For a given metric, let x_i be the performance of AttackDRO++ under seed i , and let y_i be the performance of a baseline method under the same seed. The paired difference is defined as:

$$d_i = x_i - y_i. \quad (5.1)$$

All statistical comparisons are performed on these paired differences. This paired design reduces the influence of random initialization, data shuffling, and optimization noise, allowing the analysis to focus on the relative performance difference between methods.

The primary significance test is the Wilcoxon signed-rank test. This test is selected because it is non-parametric and does not assume that the paired differences are normally distributed. This property is important for repeated-seed adversarial robustness experiments, where the number of seeds is limited and the distribution of performance differences may be irregular. For directional comparisons, the alternative hypothesis is that AttackDRO++ achieves higher performance than the baseline:

$$H_0 : \text{median}(d_i) \leq 0, \quad (5.2)$$

$$H_1 : \text{median}(d_i) > 0. \quad (5.3)$$

A paired t -test is also reported as a secondary parametric check. While the paired t -test assumes approximate normality of the paired differences, it provides a useful sanity check. When both the Wilcoxon signed-rank test and the paired t -test support the same conclusion, the result is considered more stable. If the two tests disagree, the Wilcoxon test is prioritized because it is less dependent on distributional assumptions.

In addition to p -values, this study reports the paired effect size using Cohen’s d_z :

$$d_z = \frac{\bar{d}}{s_d}, \quad (5.4)$$

where $\bar{d}$ is the mean paired difference and s_d is the sample standard deviation of the paired differences. Cohen’s d_z is used to quantify the magnitude of the improvement. Conventional thresholds are used as rough guidance, with 0.2, 0.5, and 0.8 corresponding to small, medium, and large effects, respectively. However, in adversarial robustness evaluation, even moderate effect sizes may be practically meaningful if the improvement

is consistent across strong attacks or improves worst-case robustness.

To estimate uncertainty, 95% bootstrap confidence intervals are computed from seed-level results. For each method and metric, the seed-level values are resampled with replacement 10,000 times, and the mean is recomputed for each resample. The 2.5th and 97.5th percentiles of the bootstrap distribution are used as the confidence interval. Bootstrap confidence intervals are also computed for the paired difference $x_i - y_i$, as this directly estimates the uncertainty of the improvement over each baseline.

Because multiple attacks, metrics, and baseline comparisons are evaluated, Holm–Bonferroni correction is applied within each metric family. Both raw and corrected p -values are reported, but corrected p -values are used for final significance interpretation. This correction reduces the risk of false positive findings while remaining less conservative than the standard Bonferroni correction.

The reporting format used in the results chapter is summarized in Table 5.8.

Table 5.8: Reporting format for the statistical significance protocol.

Quantity	Reported value	Purpose
Mean performance	Mean across seeds	Reports average robustness.
Seed variation	Standard deviation	Measures run-to-run variability.
Uncertainty	95% bootstrap confidence interval	Estimates uncertainty of the mean.
Primary test	Wilcoxon signed-rank test	Non-parametric paired comparison.
Secondary test	Paired t -test	Parametric sanity check.
Effect size	Cohen’s d_z	Measures magnitude of improvement.
Correction	Holm–Bonferroni correction	Controls multiple comparisons.

A comparison is considered statistically supported when AttackDRO++ improves the mean metric, obtains a corrected p -value below the significance threshold, and shows a non-trivial paired effect size. This protocol is applied consistently to white-box robustness, gray-box transfer robustness, aggregate robustness metrics, and ablation studies.

5.7 Evaluation Metrics and Robustness Definitions

This section defines the metrics used to compare all methods. Let

$$\mathcal{D}_{\text{test}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$$

denote the test set, and let A be an adversarial attack. The adversarial example generated by A is denoted by

$$\mathbf{x}_i^A = A(f_\theta, \mathbf{x}_i, y_i).$$

The robust accuracy under attack A is

$$\text{Acc}(A) = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left[\arg \max_c f_\theta(\mathbf{x}_i^A)_c = y_i \right]. \quad (5.5)$$

Clean accuracy is computed without adversarial perturbation:

$$\text{Acc}_{\text{clean}} = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left[\arg \max_c f_\theta(\mathbf{x}_i)_c = y_i \right]. \quad (5.6)$$

The primary aggregate metric is $\text{Mean}(8)$, defined as the average robust accuracy over the eight non-AutoAttack evaluation attacks:

$$\text{Mean}(8) = \frac{1}{|\mathcal{A}_8|} \sum_{A \in \mathcal{A}_8} \text{Acc}(A), \quad (5.7)$$

where

$$\mathcal{A}_8 = \{\text{FGSM-RS}, \text{PGD-}\ell_\infty, \text{TPGD}, \text{MI-FGSM}, \text{PGD-}\ell_2, \text{DDN-}\ell_2, \text{DeepFool-}\ell_2, \text{CW-}\ell_2\}.$$

We also report the worst robust accuracy across the same eight attacks:

$$\text{Worst}(8) = \min_{A \in \mathcal{A}_8} \text{Acc}(A). \quad (5.8)$$

For finer analysis, we report seen, held-out ℓ_∞ , and held-out ℓ_2 averages:

$$\text{Seen} = \frac{1}{2} [\text{Acc}(\text{PGD-}\ell_\infty) + \text{Acc}(\text{DDN-}\ell_2)], \quad (5.9)$$

$$\text{Heldout}_{\ell_\infty} = \frac{1}{3} [\text{Acc}(\text{FGSM-RS}) + \text{Acc}(\text{TPGD}) + \text{Acc}(\text{MI-FGSM})], \quad (5.10)$$

$$\text{Heldout}_{\ell_2} = \frac{1}{3} [\text{Acc}(\text{PGD-}\ell_2) + \text{Acc}(\text{DeepFool-}\ell_2) + \text{Acc}(\text{CW-}\ell_2)]. \quad (5.11)$$

AutoAttack is reported separately from $\text{Mean}(8)$. Early ablations may use AutoAttack on 512 test samples as a sanity check, while final AutoAttack comparisons are reported on the full test set for selected representative seeds.

For graybox evaluation, adversarial examples are generated using a surrogate model f_{θ_s} and evaluated on a target model f_{θ_t} :

$$\text{Acc}_{s \rightarrow t}(A) = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left[\arg \max_c f_{\theta_t} \left(A(f_{\theta_s}, \mathbf{x}_i, y_i) \right)_c = y_i \right]. \quad (5.12)$$

Thus, the same table supports whitebox, same-method transfer, and cross-method gray-box evaluation.

Finally, for fine-grained analysis, each attack-class pair is treated as an evaluation group. For attack A and class c , the attack-class group accuracy is

$$\text{Acc}(A, c) = \frac{1}{|\mathcal{D}_c|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_c} \mathbb{I} \left[\arg \max_j f_{\theta}(\mathbf{x}_i^A)_j = y_i \right], \quad (5.13)$$

where $\mathcal{D}_c$ is the subset of test examples whose label is c .

Table 5.9: Summary of evaluation metrics.

Metric	Definition	Purpose
Clean accuracy	Accuracy on unperturbed test images	Measures standard classification performance.
Per-attack robust accuracy	Accuracy under a single adversarial attack	Shows which attacks each method resists or fails against.
Mean(8)	Average accuracy over the eight non-AutoAttack attacks	Main aggregate robustness metric.
Worst(8)	Minimum accuracy over the eight non-AutoAttack attacks	Measures the weakest attack in the suite.
Seen mean	Average over PGD- ℓ_∞ and DDN- ℓ_2	Measures robustness on source attacks.
Held-out ℓ_∞ mean	Average over FGSM-RS, TPGD, and MI-FGSM	Measures transfer to unseen ℓ_∞ attacks.
Held-out ℓ_2 mean	Average over PGD- ℓ_2 , DeepFool- ℓ_2 , and CW- ℓ_2	Measures transfer to unseen ℓ_2 attacks.
AutoAttack	Standard ℓ_∞ AutoAttack robust accuracy	Separate robustness stress test.
Graybox transfer accuracy	Target accuracy on adversarial examples generated by a surrogate	Measures transfer robustness across models.
Attack-class group accuracy	Accuracy for each attack $\times$ class pair	Diagnoses fine-grained failure modes.

5.8 Graybox Transfer Protocol

To evaluate whether the proposed method changes the structure of learned robustness rather than only improving whitebox accuracy, we conduct a cross-model graybox transfer experiment. The experiment uses four method families: PGD-AT, DDN-AT, Multi-AT, and AttackDRO++ (Ours). Each method is evaluated over five representative seeds, giving 20 checkpoints in total.

Each checkpoint is used in two roles. As a *surrogate*, it generates adversarial examples. As a *target*, it is evaluated on adversarial examples generated by all surrogate checkpoints. Therefore, the full protocol evaluates

$$20 \times 20 = 400$$

surrogate–target pairs. This design produces whitebox, same-method transfer, and cross-method graybox transfer results from a single unified evaluation table.

To keep the computation tractable while still supporting per-class and per-attack analysis, each surrogate checkpoint generates a balanced adversarial cache. The CIFAR-10 test set contains 1000 images per class. For each class, the test images are split evenly across the eight evaluation attacks. Thus, each attack-class cell contains 125 source images:

$$10 \text{ classes} \times 8 \text{ attacks} \times 125 \text{ examples} = 10,000$$

adversarial examples per surrogate checkpoint.

The class-wise split is generated deterministically using a fixed shuffle seed. Therefore, the same attack-class cell always refers to the same underlying CIFAR-10 source images across surrogate checkpoints. This makes paired comparisons between surrogate and target methods well aligned.

For each surrogate–target pair, the evaluation produces one row per attack-class group, indexed by surrogate method, surrogate seed, target method, target seed, attack, and class. Since there are 80 attack-class groups per pair, the full table contains $400 \times 80 = 32,000$ rows. Transfer matrices, attack-class drilldowns, and paired method comparisons are all computed by aggregating this same long-form table. Table 5.10 summarizes the transfer regimes used in the analysis.

Table 5.10: Transfer regimes in the graybox evaluation protocol.

Regime	Condition	Rows	Purpose
Whitebox	Same method and same seed	$20 \times 80 = 1,600$	Recovers standard whitebox robustness as a sanity check.
Same-method transfer	Same method, different seed	$80 \times 80 = 6,400$	Measures seed-to-seed transfer within a method family.
Cross-method same-seed	Different methods, same seed	$60 \times 80 = 4,800$	Measures transfer between method families under matched seeds.
Cross-method cross-seed	Different methods, different seeds	$240 \times 80 = 19,200$	Measures the most general cross-method graybox transfer setting.

In Chapter 6, the cross-method same-seed and cross-method cross-seed regimes are combined to form the main graybox transfer analysis. The whitebox and same-method transfer regimes are reported separately as diagnostic checks.

Chapter 6

Results and Analysis

6.1 Whitebox Robustness Under Direct Attacks

6.1.1 Aggregate Robustness

Table 6.1: Aggregate robust accuracy (%) on the CIFAR-10 evaluation suite, reported as mean $\pm$ standard deviation across $n = 20$ seeds. AutoAttack is reported separately and is not included in Mean(8). Bold values indicate the best method for each metric.

Metric	PGD-AT	DDN-AT	Multi-AT	AttackDRO++ (Ours)
Mean(8)	60.499 $\pm$ 0.327	56.226 $\pm$ 0.233	62.045 $\pm$ 0.447	62.324 $\pm$ 0.147
Heldout(6)	62.268 $\pm$ 0.390	59.628 $\pm$ 0.254	64.083 $\pm$ 0.505	64.401 $\pm$ 0.155
Heldout ℓ_∞ (3)	62.515 $\pm$ 0.334	51.594 $\pm$ 0.306	60.916 $\pm$ 0.427	61.149 $\pm$ 0.170
Heldout ℓ_2 (3)	62.021 $\pm$ 0.474	67.661 $\pm$ 0.278	67.249 $\pm$ 0.603	67.654 $\pm$ 0.223
Seen mean (2)	55.192 $\pm$ 0.182	46.020 $\pm$ 0.247	55.934 $\pm$ 0.303	56.093 $\pm$ 0.166
Worst(8)	49.656 $\pm$ 0.161	25.957 $\pm$ 0.378	45.082 $\pm$ 0.281	45.079 $\pm$ 0.237

We compare four methods on $n = 20$ paired training seeds: PGD-AT, DDN-AT, Multi-AT, and the proposed AttackDRO++ (Anchor35 GradFP). AttackDRO++ improves Mean(8) over Multi-AT by +0.279 pp ($p = 0.0076$, 14/20 wins), with the strongest channel on heldout ℓ_2 (+0.405 pp, $p = 0.0079$, 16/20). It improves Mean(8) over PGD-AT by +1.826 pp and over DDN-AT by +6.098 pp, both at $p < 10^{-10}$ on all 20 seeds, and achieves 2.5–3.3 $\times$ tighter seed-to-seed standard deviation than Multi-AT on every average-case metric. Worst(8) and AutoAttack are statistically indistinguishable from Multi-AT ($|\Delta| < 0.005$ pp, $p > 0.97$): AttackDRO++ improves average and held-

out robustness without disturbing standardized worst-case behavior. The single-attack baselines remain stronger than AttackDRO++ on the family they train against (PGD-AT on heldout ℓ_∞ , DDN-AT on heldout ℓ_2) but lose substantially on the other, confirming the specialist-versus-balanced interpretation that motivates Multi-ATtack training.

AttackDRO++ achieves the highest Mean(8), Heldout(6), and Seen mean. PGD-AT wins Heldout ℓ_∞ and Worst(8); DDN-AT wins Heldout ℓ_2 . The bold pattern visualizes the specialist-versus-balanced interpretation: each single-AT method dominates one column, while AttackDRO++ is best on every column-aggregating metric.

Effect-size summary. Table 6.2 reports paired effect magnitudes between AttackDRO++ and each baseline. Cohen’s $d_z = \Delta/\sigma_\Delta$ is the standardized paired effect size; conventional thresholds are 0.2 (small), 0.5 (medium), 0.8 (large). Statistical significance is reported in Table 6.3.

Table 6.2: Effect-size summary: AttackDRO++ (Ours) vs. each baseline ($n = 20$). Δ in percentage points; negative values mean the baseline is stronger.

Metric	vs. Multi-AT		vs. PGD-AT		vs. DDN-AT	
	Δ (pp)	d_z	Δ (pp)	d_z	Δ (pp)	d_z
Mean(8)	+0.279	0.67	+1.826	5.98	+6.098	23.09
Heldout(6)	+0.319	0.67	+2.134	5.65	+4.774	16.91
Heldout ℓ_∞	+0.233	0.57	−1.366	−4.27	+9.555	25.44
Heldout ℓ_2	+0.405	0.66	+5.633	11.64	−0.008	−0.03
Seen mean	+0.159	0.55	+0.901	4.59	+10.073	31.72
Worst(8)	−0.003	−0.01	−4.576	−15.36	+19.122	36.91

Two regimes are visible. Against Multi-AT, every average-case metric is a medium-effect gain ($d_z \in [0.55, 0.67]$, $\Delta \in [+0.16, +0.41]$ pp) with Worst(8) at zero effect: AttackDRO++ shifts the average-case distribution by a small but consistent amount without disturbing worst-case behavior. Against the single-AT baselines, d_z exceeds 4 on most metrics but flips negative on the family each specialist trains against. The medium-effect Multi-AT comparison validates the cluster-DRO objective as a refinement of Multi-ATtack training; the large-but- asymmetric single-AT comparisons quantify the specialist tradeoff that motivates the Multi-ATtack setting in the first place.

Paired aggregate robustness comparisons. Table 6.3 reports paired aggregate results separate the main Multi-ATtack comparison from the single-attack specialist comparisons. AutoAttack is excluded from this table and reported separately in Section 6.1.3.

Table 6.3: Paired aggregate robustness comparison between AttackDRO++ (Ours) and each baseline. Δ is computed as AttackDRO++ minus the corresponding baseline in percentage points. The p -value is from a two-sided paired t -test.

Baseline	Metric	n	Δ (pp)	σ_Δ	Wins	d_z	p
<i>AttackDRO++ (Ours) vs. Multi-AT</i>							
Multi-AT	Mean(8)	20	+0.279	0.417	14/20	0.668	0.0076
Multi-AT	Heldout(6)	20	+0.319	0.478	13/20	0.667	0.0076
Multi-AT	Heldout ℓ_∞	20	+0.233	0.412	14/20	0.566	0.0204
Multi-AT	Heldout ℓ_2	20	+0.405	0.609	16/20	0.664	0.0079
Multi-AT	Seen mean	20	+0.159	0.291	15/20	0.546	0.0246
Multi-AT	Worst(8)	20	−0.003	0.369	9/20	−0.008	0.971
<i>AttackDRO++ (Ours) vs. PGD-AT</i>							
PGD-AT	Mean(8)	20	+1.826	0.305	20/20	5.980	$< 10^{-10}$
PGD-AT	Heldout(6)	20	+2.134	0.378	20/20	5.650	$< 10^{-10}$
PGD-AT	Heldout ℓ_∞	20	−1.366	0.320	0/20	−4.273	$< 10^{-10}$
PGD-AT	Heldout ℓ_2	20	+5.633	0.484	20/20	11.638	$< 10^{-20}$
PGD-AT	Seen mean	20	+0.901	0.197	20/20	4.585	$< 10^{-10}$
PGD-AT	Worst(8)	20	−4.576	0.298	0/20	−15.364	$< 10^{-20}$
<i>AttackDRO++ (Ours) vs. DDN-AT</i>							
DDN-AT	Mean(8)	20	+6.098	0.264	20/20	23.087	$< 10^{-20}$
DDN-AT	Heldout(6)	20	+4.774	0.282	20/20	16.910	$< 10^{-20}$
DDN-AT	Heldout ℓ_∞	20	+9.555	0.376	20/20	25.440	$< 10^{-20}$
DDN-AT	Heldout ℓ_2	20	−0.008	0.319	10/20	−0.025	0.914
DDN-AT	Seen mean	20	+10.073	0.318	20/20	31.723	$< 10^{-20}$
DDN-AT	Worst(8)	20	+19.122	0.518	20/20	36.909	$< 10^{-20}$

Against Multi-AT, AttackDRO++ gives small but statistically supported gains on all average-case metrics, while Worst(8) is unchanged. Against PGD-AT, AttackDRO++ improves Mean(8) and held-out ℓ_2 robustness, but PGD-AT remains stronger on held-out ℓ_∞ and Worst(8). Against DDN-AT, AttackDRO++ improves all ℓ_∞ -related aggregates while matching DDN-AT on held-out ℓ_2 . This again supports the main interpretation that single-attack baselines are specialists, while AttackDRO++ provides better average cross-attack balance.

Table 6.4: Multiple-comparison-corrected paired significance tests for AttackDRO++ (Ours) against each baseline across $n = 20$ paired seeds. Δ is computed as AttackDRO++ minus the corresponding baseline in percentage points. The table reports bootstrap 95% confidence intervals, seed-level win counts, Cohen’s paired effect size d_z , paired t -test p -values, Wilcoxon signed-rank p -values, and Holm-corrected Wilcoxon p -values. Holm correction is applied within each baseline comparison across the seven reported metrics. AutoAttack is excluded from this table and reported separately in Section 6.1.3.

Baseline	Metric	Δ (pp)	95% CI (pp)	Wins	d_z	p_t	p_W	$p_{W, \text{Holm}}$
<i>AttackDRO++ (Ours) vs. Multi-AT</i>								
Multi-AT	Clean	+0.539	[+0.195, +0.903]	13/20	0.642	0.0098	0.0441	0.1449
Multi-AT	Mean(8)	+0.279	[+0.108, +0.463]	14/20	0.668	0.0076	0.0136	0.0953
Multi-AT	Heldout(6)	+0.319	[+0.123, +0.527]	13/20	0.667	0.0076	0.0192	0.0962
Multi-AT	Heldout ℓ_∞	+0.233	[+0.060, +0.410]	14/20	0.566	0.0204	0.0362	0.1449
Multi-AT	Heldout ℓ_2	+0.405	[+0.144, +0.661]	16/20	0.664	0.0079	0.0136	0.0953
Multi-AT	Seen mean	+0.159	[+0.038, +0.285]	15/20	0.546	0.0246	0.0419	0.1449
Multi-AT	Worst(8)	-0.003	[-0.168, +0.155]	9/20	-0.008	0.9714	0.9839	0.9839
<i>AttackDRO++ (Ours) vs. PGD-AT</i>								
PGD-AT	Clean	+4.015	[+3.666, +4.386]	20/20	4.779	< 0.0001	< 0.0001	< 0.0001
PGD-AT	Mean(8)	+1.826	[+1.702, +1.965]	20/20	5.980	< 0.0001	< 0.0001	< 0.0001
PGD-AT	Heldout(6)	+2.134	[+1.980, +2.303]	20/20	5.650	< 0.0001	< 0.0001	< 0.0001
PGD-AT	Heldout ℓ_∞	-1.366	[-1.496, -1.223]	0/20	-4.273	< 0.0001	< 0.0001	< 0.0001
PGD-AT	Heldout ℓ_2	+5.633	[+5.441, +5.849]	20/20	11.638	< 0.0001	< 0.0001	< 0.0001
PGD-AT	Seen mean	+0.901	[+0.815, +0.984]	20/20	4.585	< 0.0001	< 0.0001	0.0002
PGD-AT	Worst(8)	-4.576	[-4.708, -4.454]	0/20	-15.364	< 0.0001	< 0.0001	0.0002
<i>AttackDRO++ (Ours) vs. DDN-AT</i>								
DDN-AT	Clean	-4.563	[-4.712, -4.414]	0/20	-13.134	< 0.0001	< 0.0001	< 0.0001
DDN-AT	Mean(8)	+6.098	[+5.986, +6.209]	20/20	23.087	< 0.0001	< 0.0001	< 0.0001
DDN-AT	Heldout(6)	+4.774	[+4.654, +4.893]	20/20	16.910	< 0.0001	< 0.0001	< 0.0001
DDN-AT	Heldout ℓ_∞	+9.555	[+9.399, +9.718]	20/20	25.440	< 0.0001	< 0.0001	< 0.0001
DDN-AT	Heldout ℓ_2	-0.008	[-0.143, +0.127]	10/20	-0.025	0.9137	0.9563	0.9563
DDN-AT	Seen mean	+10.072	[+9.938, +10.208]	20/20	31.723	< 0.0001	< 0.0001	0.0002
DDN-AT	Worst(8)	+19.122	[+18.904, +19.344]	20/20	36.909	< 0.0001	< 0.0001	< 0.0001

Table 6.3 reports the paired t -test comparison. To align with the statistical protocol described in Chapter 5, Table 6.4 additionally reports Wilcoxon signed-rank tests with Holm correction, bootstrap confidence intervals, paired effect sizes, and seed-level win counts. The multiple-comparison-corrected tests provide a conservative check on the

paired t -test results. Against Multi-AT, the key question is whether the small positive gains on average-case metrics remain supported after Wilcoxon testing and Holm correction. Clean accuracy is included as a sanity metric, but the main robustness claims are based on Mean(8), heldout metrics, seen mean, and Worst(8). Against the single-attack baselines, the same analysis mainly verifies the specialist-versus-balanced pattern: PGD-AT remains stronger on ℓ_∞ -dominated metrics, DDN-AT remains competitive on heldout ℓ_2 , and AttackDRO++ provides stronger average cross-attack balance.

After Holm correction, the Multi-AT comparison should be interpreted cautiously. AttackDRO++ shows small positive differences over Multi-AT on Clean, Mean(8), Heldout(6), Heldout ℓ_∞ , Heldout ℓ_2 , and Seen mean, and these differences are supported by the paired t -tests, raw Wilcoxon tests, confidence intervals, and moderate paired effect sizes. However, the Holm-corrected Wilcoxon p -values for these metrics do not remain below the 0.05 threshold. Therefore, relative to Multi-AT, the evidence is best described as a small positive trend rather than a uniformly correction-robust improvement. Worst(8) is essentially unchanged, indicating that AttackDRO++ preserves the worst-case attack performance of Multi-AT rather than improving it.

For the single-attack baselines, the corrected tests support the expected specialist-versus-balanced interpretation. Compared with PGD-AT, AttackDRO++ achieves correction-robust gains on Mean(8), Heldout(6), Heldout ℓ_2 , and Seen mean, but remains weaker on Heldout ℓ_∞ and Worst(8), consistent with PGD-AT’s specialization toward ℓ_∞ robustness. Compared with DDN-AT, AttackDRO++ obtains correction-robust gains on Mean(8), Heldout(6), Heldout ℓ_∞ , Seen mean, and Worst(8), while Heldout ℓ_2 is statistically indistinguishable. Overall, Table 6.4 suggests that AttackDRO++ mainly improves average cross-attack balance and heldout robustness, while its worst-case behavior depends on the baseline being compared.

6.1.2 Per-Attack Robustness Breakdown

The aggregate metrics show the overall trend, but they do not reveal which attacks drive the improvement. Table 6.5 reports per-attack paired comparisons between AttackDRO++ and each baseline. The table is divided into three panels: the main comparison against Multi-AT, followed by the two single-attack specialist comparisons.

Table 6.5: Per-attack robust accuracy (%) for all four methods, $n = 20$ seeds. Bold cells indicate the highest accuracy for each attack. The right-hand Δ and p columns report the paired comparison of AttackDRO++ (Ours) against Multi-AT. AttackDRO++ (Ours) wins the highest score on 4 of 8 attacks; the two single-attack baselines together account for the remaining 4, each winning on the family they were trained against. Δ values use the bold-marker convention * for $p < 0.05$ and ** for $p < 0.01$.

Attack	Family	PGD-AT	DDN-AT	Multi-AT	AttackDRO++ (Ours)	Δ vs. Multi-AT	p
FGSM-RS	heldout ℓ_∞	67.896	64.834	68.483	68.939	+0.456**	0.0034
PGD- ℓ_∞	seen ℓ_∞	49.656	25.957	45.082	45.079	-0.003	0.971
TPGD	heldout ℓ_∞	68.388	59.310	66.934	67.043	+0.109	0.513
MI-FGSM	heldout ℓ_∞	51.260	30.638	47.331	47.466	+0.135	0.116
PGD- ℓ_2	heldout ℓ_2	64.319	69.581	69.197	69.571	+0.374*	0.0120
DDN- ℓ_2	seen ℓ_2	60.728	66.083	66.785	67.107	+0.321*	0.0136
DeepFool- ℓ_2	heldout ℓ_2	62.533	68.290	67.301	67.805	+0.504**	0.0082
CW- ℓ_2	heldout ℓ_2	59.211	65.113	65.249	65.585	+0.336**	0.0097

Table 6.5 disaggregates the Mean(8) result into the eight evaluation attacks. AttackDRO++ achieves the highest per-attack accuracy on four of eight attacks, including all three heldout ℓ_2 attacks where it beats Multi-AT at $p < 0.05$. PGD-AT wins on the three heldout ℓ_∞ attacks and on the seen attack PGD- ℓ_∞ that it trains directly against; DDN-AT wins on PGD- ℓ_2 and DeepFool- ℓ_2 , the heldout ℓ_2 attacks closest to its training distribution. The pattern is consistent with a specialist interpretation of single-attack training: each single-AT method dominates a narrow attack family at the cost of substantial regression on the other. AttackDRO++’s improvement over Multi-AT is largest on heldout ℓ_2 attacks (+0.374 to +0.504 pp, all $p < 0.05$) and smallest on the seen ℓ_∞ attack (-0.003 pp), which explains why Worst(8) is preserved rather than improved.

6.1.3 Worst-Case Behavior: Worst(8) and AutoAttack

Two metrics test standardized worst-case behavior. Worst(8) is the minimum robust accuracy across the eight evaluation attacks per checkpoint; AutoAttack- ℓ_∞ is an ensemble ℓ_∞ stress test following Croce and Hein [3]. Tables 6.6 and 6.7 report each metric separately because they have different evaluation panels: Worst(8) is computed across the full $n = 20$ seed panel, while AutoAttack is reported in two layers — a 512-sample sanity evaluation across all $n = 20$ seeds and a full-test evaluation on the selected $n = 5$ seed panel.

Table 6.6: Worst(8) and 512-sample AutoAttack sanity results across $n = 20$ seeds. Worst(8) is the minimum robust accuracy over the eight non-AutoAttack evaluation attacks. For AttackDRO++ – Multi-AT, the paired differences are -0.003 pp on Worst(8) ($p = 0.971$) and -0.001 pp on AutoAttack ($p = 0.998$). Bold values indicate the highest method.

Method	Worst(8)		AutoAttack- ℓ_∞	
	Mean (%)	σ (pp)	Mean (%)	σ (pp)
PGD-AT	49.656	0.161	44.912	1.136
DDN-AT	25.957	0.378	23.809	0.796
Multi-AT	45.082	0.281	41.318	1.167
AttackDRO++ (Ours)	45.079	0.237	41.317	1.050

Table 6.7: Full-test AutoAttack- ℓ_∞ results on the selected $n = 5$ paired seed panel. The paired difference is AttackDRO++ – Multi-AT: -0.006 pp, with $p = 0.9584$.

Method	Mean (%)	σ (pp)
Multi-AT	42.47	0.30
AttackDRO++ (Ours)	42.46	0.20

AttackDRO++ vs. Multi-AT. AttackDRO++ is statistically indistinguishable from Multi-AT on both worst-case metrics. On the $n = 20$ sanity panel, the paired differences are -0.003 pp for Worst(8) ($p = 0.971$) and -0.001 pp for AutoAttack ($p = 0.998$). The full-test AutoAttack panel gives the same conclusion (-0.006 pp, $p = 0.9584$). Thus, the claim is preservation rather than improvement: AttackDRO++ improves average and heldout robustness without degrading standardized worst-case behavior.

PGD-AT specialization. PGD-AT obtains the highest Worst(8) and AutoAttack because both metrics are dominated by ℓ_∞ robustness, the threat family it trains against. This advantage is therefore structural rather than evidence of broader robustness. As Table 6.5 shows, PGD-AT loses 5–6 pp to AttackDRO++ on heldout ℓ_2 attacks, while DDN-AT shows the opposite failure mode by collapsing under ℓ_∞ stress. Single-attack training therefore produces specialists; AttackDRO++ does not match PGD-AT’s ℓ_∞ depth, but provides the best average cross-attack balance.

6.1.4 Variance Reduction and Training Stability

In addition to improving the mean of average-case robustness metrics, AttackDRO++ also reduces seed-to-seed variation relative to Multi-AT. Table 6.8 compares the standard deviation of each seed-level metric between the two methods.

Table 6.8: Variance reduction across $n = 20$ training seeds. σ is the standard deviation of the per-seed metric. AttackDRO++ substantially reduces seed-to-seed variation on Mean(8), held-out averages, and held-out norm-specific metrics, while Worst(8) and AutoAttack standard deviations remain similar across methods.

Metric	σ Multi-AT (pp)	σ AttackDRO++ (pp)	Ratio
Mean(8)	0.447	0.148	3.03 $\times$
Heldout(6)	0.505	0.155	3.26 $\times$
Heldout ℓ_2	0.603	0.223	2.70 $\times$
Heldout ℓ_∞	0.427	0.170	2.51 $\times$
Seen mean	0.303	0.166	1.83 $\times$
Worst(8)	0.281	0.237	1.18 $\times$
AutoAttack- ℓ_∞	1.167	1.050	1.11 $\times$

The reduction is strongest for held-out robustness. The standard deviation decreases by $3.26\times$ for Heldout(6), $2.70\times$ for held-out ℓ_2 , and $2.51\times$ for held-out ℓ_∞ . Mean(8) also becomes substantially more stable, decreasing from 0.447 pp under Multi-AT to 0.148 pp under AttackDRO++, a $3.03\times$ reduction.

This suggests that the anchored Cluster-DRO objective not only improves the mean of average-case robustness metrics, but also makes the Multi-AT attack training outcome less sensitive to random seed. The effect is weaker for Worst(8) and AutoAttack, whose standard deviations remain similar across methods. This is consistent with the previous sections: AttackDRO++ preserves the hardest ℓ_∞ stress metrics relative to Multi-AT, while its main benefit is on average and held-out robustness.

6.2 Class-wise Whitebox Robustness Across Attacks

The aggregate results in Section 6.1 show that AttackDRO++ improves average cross-attack robustness while preserving a similar worst-case profile to the uniform multi-attack baseline. However, aggregate accuracy can hide localized failure modes: a method may improve the mean by increasing already-easy cells, while leaving hard class–attack combinations unchanged. To examine whether the improvement is distributed across the evaluation grid, we perform a whitebox per-(attack $\times$ class) drilldown on the eight non-AutoAttack attacks and the ten CIFAR-10 classes. Each cell therefore corresponds to the robust accuracy of one method on one attack and one class, averaged over five seeds.

This analysis has two purposes. First, it identifies whether AttackDRO++ behaves as a balanced multi-attack method or merely shifts robustness from one threat family to another. Second, it exposes the tail cells that dominate practical failure behavior, especially under hard attacks such as PGD- ℓ_∞ and MI-FGSM on visually ambiguous classes. We compare AttackDRO++ against three baselines: the uniform multi-attack model (Multi-AT), the PGD- ℓ_∞ specialist (PGD-AT), and the DDN- ℓ_2 specialist (DDN-AT).

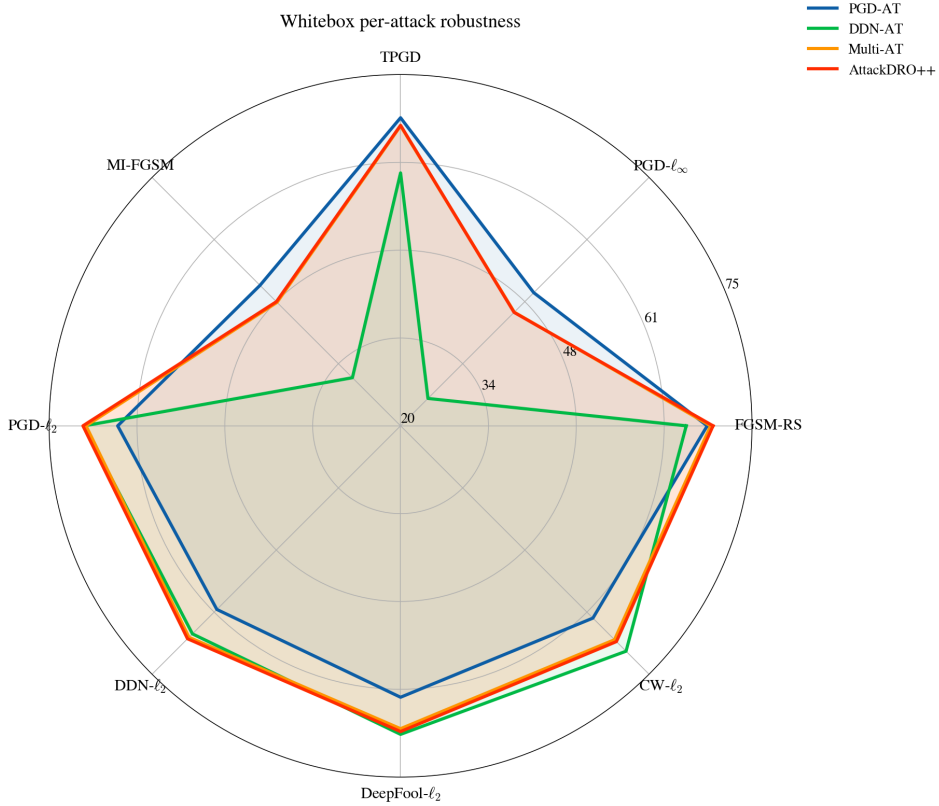


Figure 6.1: Whitebox per-attack robustness profile averaged over five seeds. The radar plot compares PGD-AT, DDN-AT, Multi-AT, and AttackDRO++ (Ours) across the eight non-AutoAttack evaluation attacks. The shape of each curve highlights the specialization of the single-attack baselines and the more balanced profile of the multi-attack methods.

Absolute per-class heatmaps with a shared scale. Before analyzing deltas, we first report the raw per-(class $\times$ attack) robust accuracy of each method. All four heatmaps use the same global color scale from 0% to 100%. This shared scale is important because it makes the maps visually comparable across methods: a darker cell always represents the same absolute robust accuracy, regardless of which method is being shown.

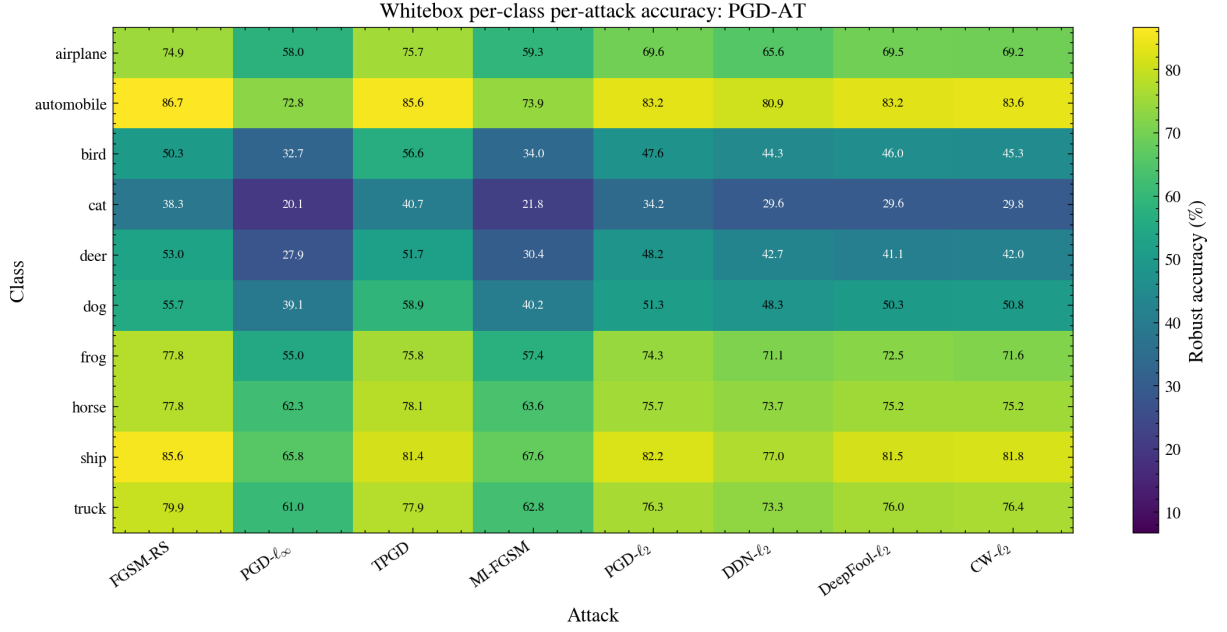


Figure 6.2: Absolute whitebox per-(class $\times$ attack) robust accuracy for PGD-AT. Cells report robust accuracy (%).

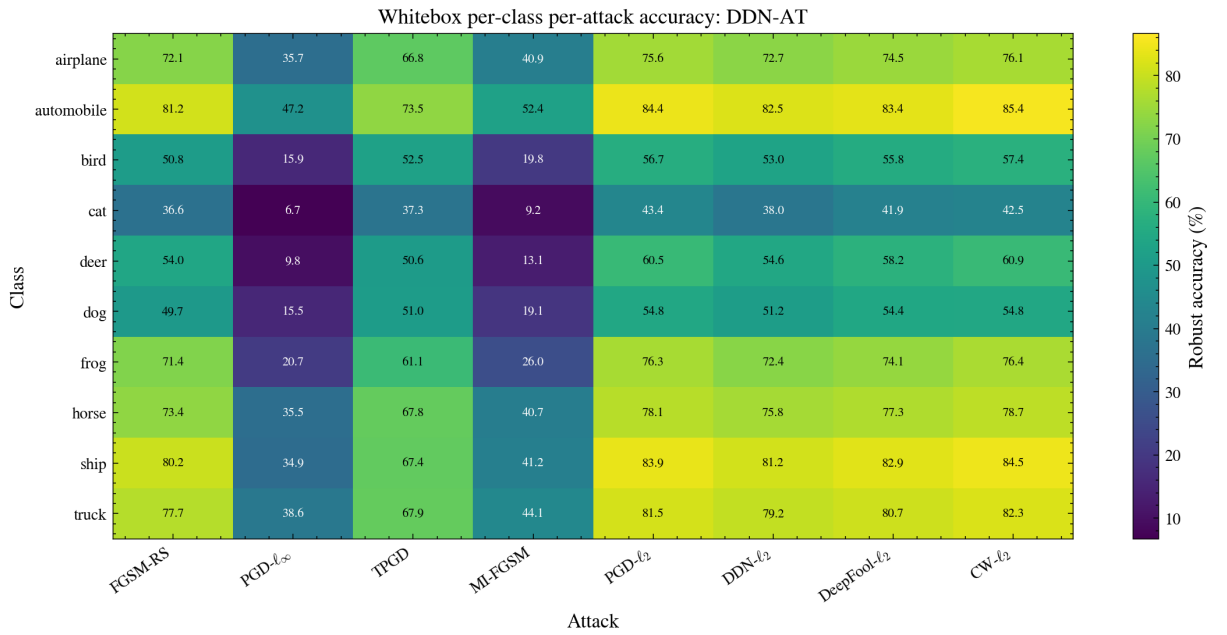


Figure 6.3: Absolute whitebox per-(class $\times$ attack) robust accuracy for DDN-AT. Cells report robust accuracy (%).

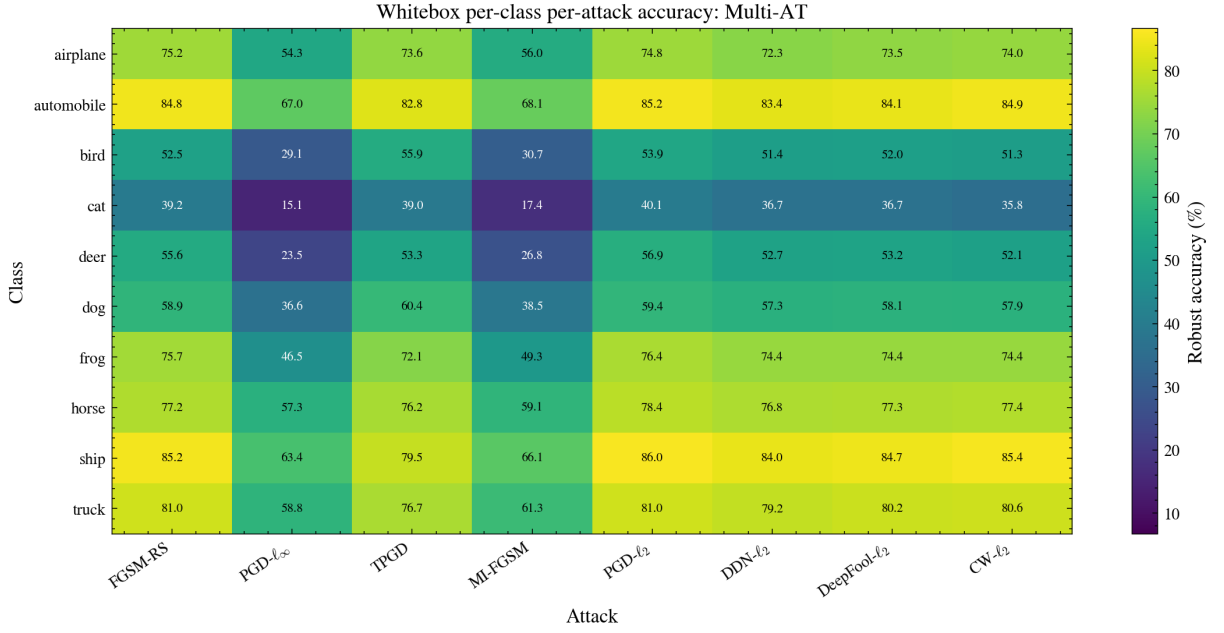


Figure 6.4: Absolute whitebox per-(class $\times$ attack) robust accuracy for Multi-AT. Cells report robust accuracy (%).

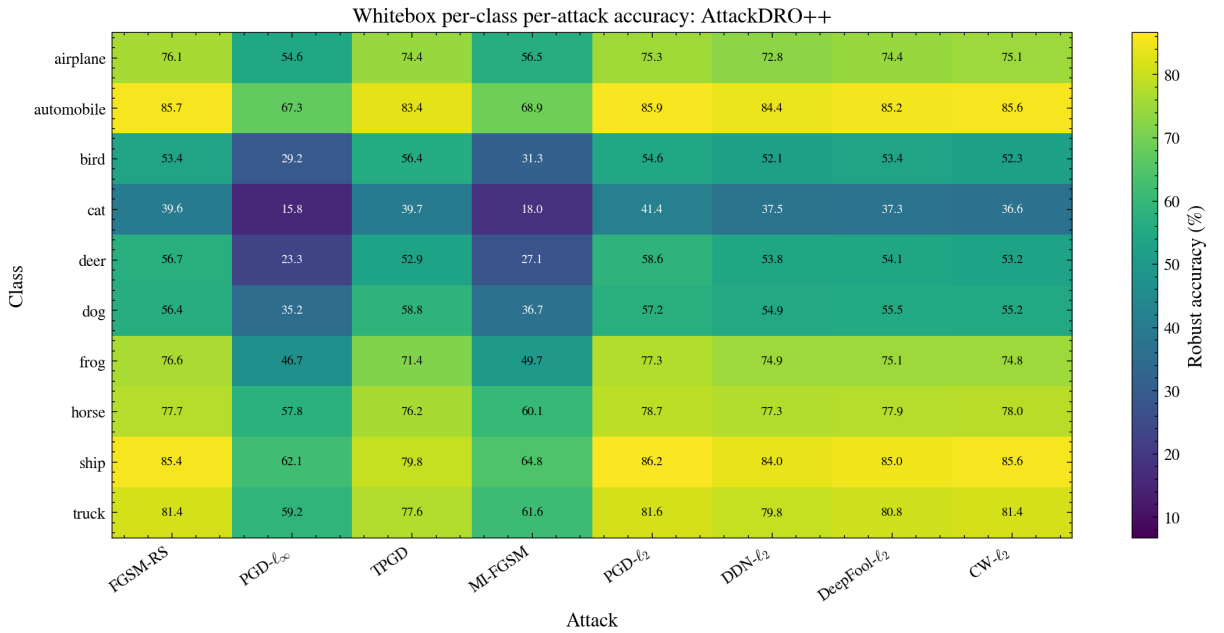


Figure 6.5: Absolute whitebox per-(class $\times$ attack) robust accuracy for AttackDRO++ (Ours). Cells report robust accuracy (%).

PGD-AT is stronger on several ℓ_∞ cells, while DDN-AT is more biased toward bounded ℓ_2 attacks and visibly weaker on PGD- ℓ_∞ and MI-FGSM. Compared with the specialists, Multi-AT and AttackDRO++ show a more balanced cross-attack profile. AttackDRO++ largely preserves the Multi-AT structure while slightly lifting many cells, especially outside the hardest ℓ_∞ animal-class tail.

6.2.1 Per-Class Whitebox Robustness Pattern

Across all methods, the per-class robustness pattern is highly non-uniform. Classes such as *automobile*, *ship*, and *truck* tend to remain easier under most attacks, while *cat*, *deer*, *bird*, and *dog* form the recurring hard tail. This pattern is consistent with the visual similarity and intra-class variation of CIFAR-10: animal classes often share local textures and backgrounds, so small adversarial perturbations can more easily move samples across nearby decision regions.

The hardest whitebox cells are concentrated in the ℓ_∞ iterative attacks. For AttackDRO++, the lowest cells are PGD20-CE on *cat* and *deer*, followed by MI-FGSM on the same classes. This indicates that, even though AttackDRO++ improves the average multi-attack profile, the most difficult failure modes are still driven by strong ℓ_∞ optimization on semantically similar animal classes. In contrast, bounded ℓ_2 attacks such as PGD- ℓ_2 , DDN- ℓ_2 , DeepFool- ℓ_2 , and CW- ℓ_2 are generally less damaging for the proposed model than the strongest ℓ_∞ cells.

Attack-Class Delta Against Multi-AT Baseline Compared with Multi-AT, AttackDRO++ produces a broad but modest shift rather than a large change concentrated in a few cells. The mean per-cell gain is only +0.281 percentage points, but 67 out of 80 attack–class cells are positive.



Figure 6.6: Whitebox per-(class $\times$ attack) delta of AttackDRO++ relative to Multi-AT. Each cell reports AttackDRO++ minus Multi-AT in percentage points, averaged over five seeds.

Attack-Class Delta Against Single-AT Baselines Table 6.9 summarizes the per-cell delta between AttackDRO++ and each baseline. The comparison against Multi-AT is the most important one because both methods train on the same source attacks and differ mainly in the group structure used for optimization. AttackDRO++ improves 67/80 cells over Multi-AT, with a mean gain of +0.281 percentage points and a bottom-10 lift of +0.282 percentage points. The gains are small, but they are widespread, which supports the interpretation that cluster-based DRO acts as a stabilizing refinement over uniform multi-attack training.

Table 6.9: Per-(class $\times$ attack) whitebox comparison of AttackDRO++ against each baseline. Δ is computed as AttackDRO++ minus the baseline. Positive cells count the number of attack–class cells where $\Delta > 0$ out of 80 total cells.

Baseline	Positive cells ($\Delta > 0$)	Mean Δ (pp)	Significant cells AttackDRO++ / Baseline	Bottom-10 lift (pp)	Main pattern
Multi-AT	67/80	+0.281	0 / 2	+0.282	Balanced refinement
PGD-AT	47/80	+1.782	40 / 19	+0.726	Gains mainly on ℓ_2 cells
DDN-AT	62/80	+5.781	44 / 5	+15.710	Gains mainly on ℓ_∞ cells

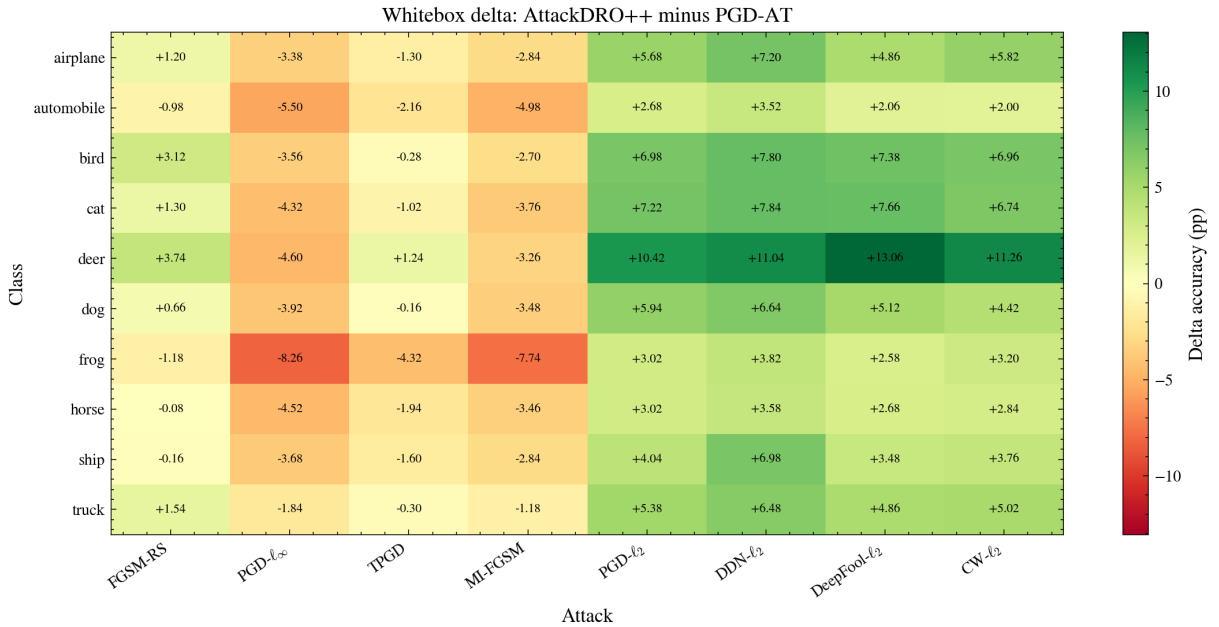


Figure 6.7: Whitebox per-(class $\times$ attack) delta of AttackDRO++ (Ours) relative to PGD-AT. Positive cells indicate where AttackDRO++ (Ours) improves over the PGD- ℓ_∞ specialist. The figure shows that AttackDRO++ (Ours) gains mainly on bounded ℓ_2 attacks while PGD-AT remains stronger on several ℓ_∞ iterative cells.

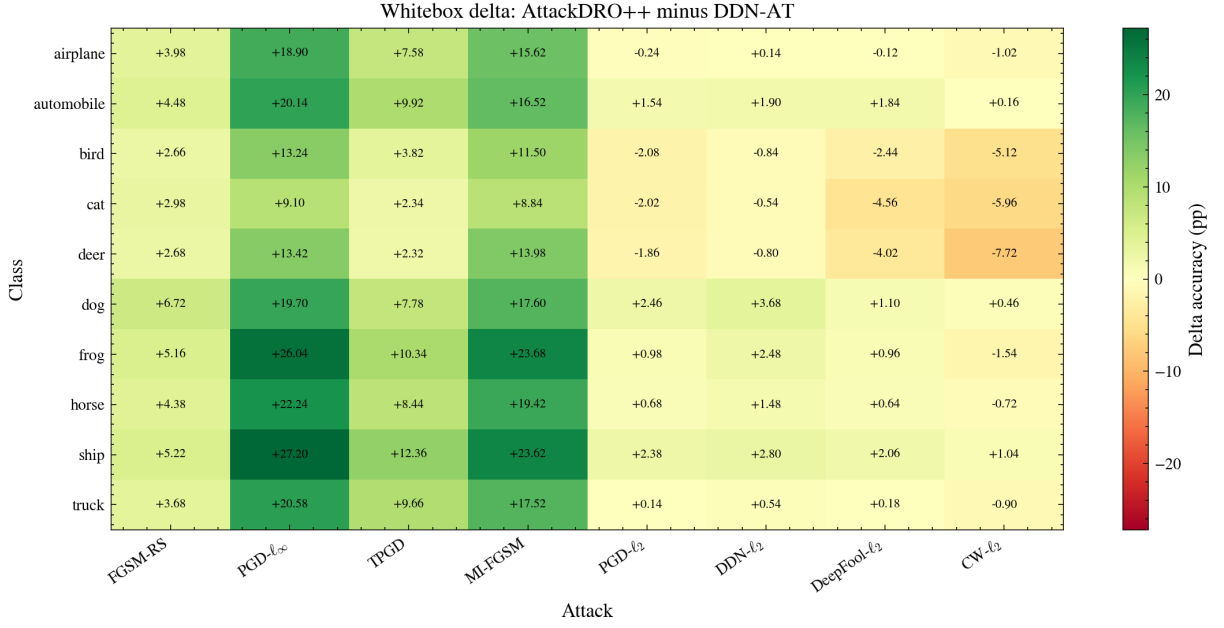


Figure 6.8: Whitebox per-(class $\times$ attack) delta of AttackDRO++ (Ours) relative to DDN-AT. The large positive cells under PGD- ℓ_∞ and MI-FGSM show that AttackDRO++ (Ours) substantially corrects the ℓ_∞ weakness of the DDN- ℓ_2 specialist, while several negative cells remain on DDN-AT’s preferred ℓ_2 family.

The comparison against PGD-AT shows the expected specialist trade-off. AttackDRO++ is positive in 47/80 cells and has a mean gain of +1.782 percentage points, but the gains and losses are not uniformly distributed. The largest improvements occur on bounded ℓ_2 attacks, especially DeepFool- ℓ_2 , CW- ℓ_2 , DDN- ℓ_2 , and PGD- ℓ_2 on hard animal classes such as *deer*, *cat*, and *bird*. For example, the largest gains over PGD-AT appear on DeepFool- ℓ_2 for *deer* (+13.06 pp), CW- ℓ_2 for *deer* (+11.26 pp), and DDN- ℓ_2 for *deer* (+11.04 pp). Conversely, PGD-AT remains stronger on several ℓ_∞ cells, particularly PGD20-CE and MI-FGSM for *frog*, *automobile*, *horse*, and *cat*. This confirms that PGD-AT is not weak in absolute terms; rather, it is specialized toward the ℓ_∞ geometry used during training.

The comparison against DDN-AT is even more asymmetric. AttackDRO++ improves 62/80 cells and gains +5.781 percentage points on average, with a large +15.710 percentage-point improvement on the bottom-10 cells. The largest improvements occur under ℓ_∞ attacks: PGD20-CE on *ship* (+27.20 pp), PGD20-CE on *frog* (+26.04 pp), MI-FGSM on *frog* (+23.68 pp), and MI-FGSM on *ship* (+23.62 pp). These large deltas show that DDN-AT develops a strong ℓ_2 bias and leaves many ℓ_∞ attack–class cells under-protected. At the same time, DDN-AT remains stronger on some ℓ_2 boundary/optimization cells, especially CW- ℓ_2 and DeepFool- ℓ_2 for *deer*, *cat*, and *bird*. Thus, AttackDRO++ improves cross-attack balance but does not completely dominate the

strongest single-norm specialist on its preferred family.

6.2.2 Attack-Family Interpretation

The attack-wise deltas make the specialization effect clear. Relative to PGD-AT, AttackDRO++ loses on PGD20-CE and MI-FGSM on average, but gains strongly on the bounded ℓ_2 attacks. Relative to DDN-AT, the pattern reverses: AttackDRO++ gains substantially on PGD20-CE, MI-FGSM, TPGD, and FGSM-RS, while being slightly weaker on CW- ℓ_2 and DeepFool- ℓ_2 . The Multi-AT comparison is more balanced: most attack averages are positive, but the magnitudes are below one percentage point.

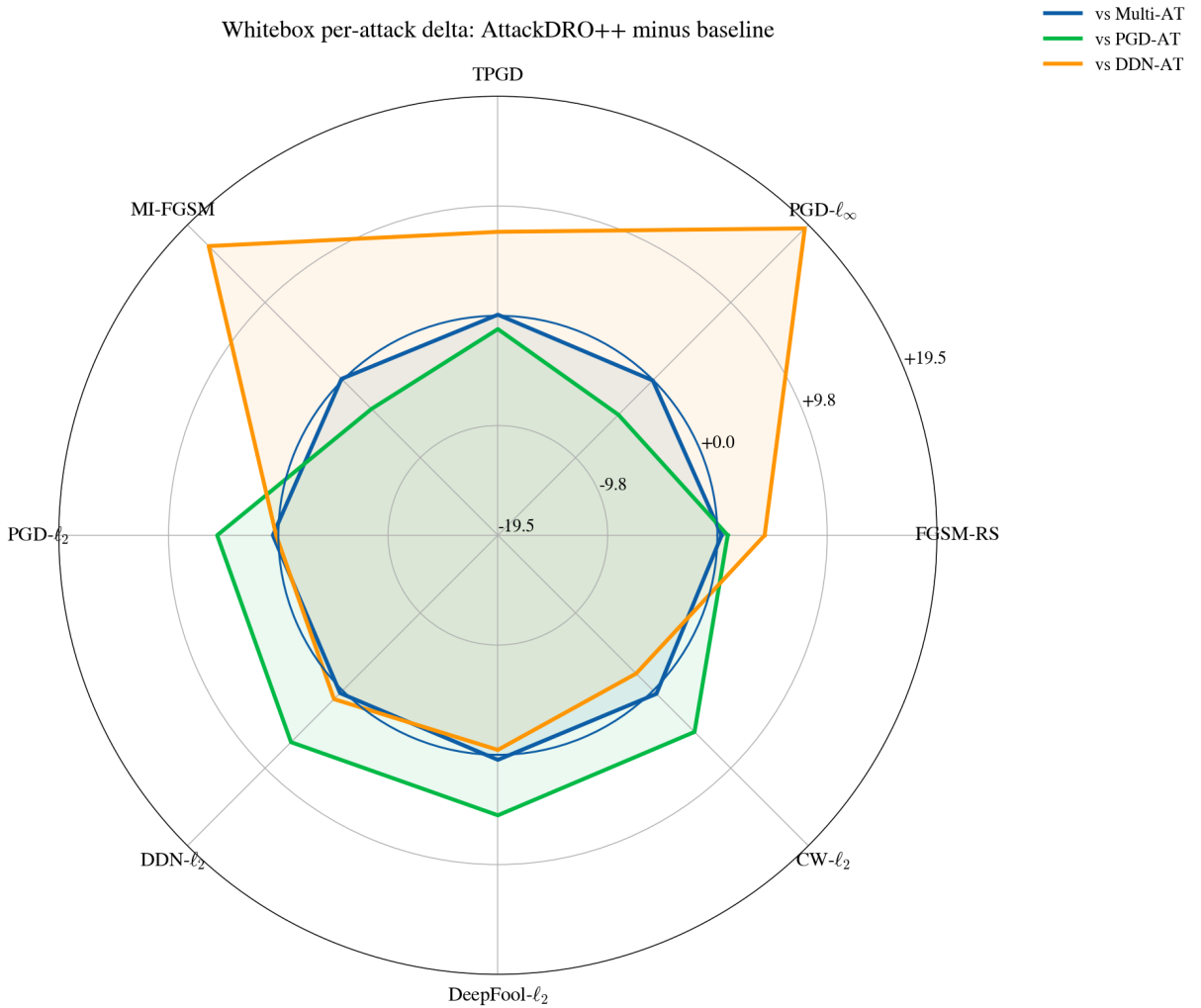


Figure 6.9: Attack-wise whitebox delta of AttackDRO++ relative to the three baselines.

This result supports the main claim of the chapter. AttackDRO++ should not be interpreted as a replacement for a single specialist when the deployment threat model is known exactly. Instead, it is a cross-attack robustness method: it sacrifices some of the peak strength of a specialist in exchange for a more even robustness profile across attacks and classes.

6.2.3 Bottom- K Whitebox Tail Robustness

We next examine the weakest attack–class cells for each baseline, using $K \in \{5, 10, 20\}$. These cells capture localized failure modes that can be hidden by mean accuracy. For Multi-AT, the weakest cells are mainly strong ℓ_∞ attacks, especially PGD20-CE and MI-FGSM, on hard animal classes such as *cat*, *deer*, and *bird*. AttackDRO++ improves this tail only slightly: the bottom-10 lift is +0.282 pp. Although small, this shows that the average gain over Multi-AT is not obtained by sacrificing its weakest cells.

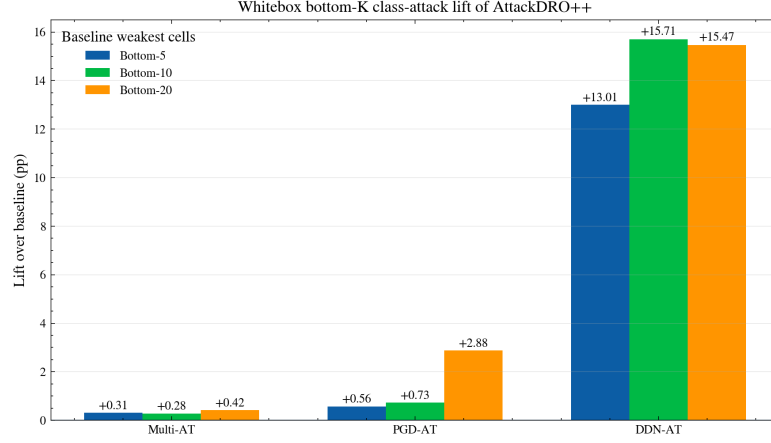


Figure 6.10: Bottom- K whitebox class–attack lift of AttackDRO++ over each baseline. For each baseline, the bottom- K cells are the weakest class–attack cells of that baseline; bars report AttackDRO++ minus the baseline on those same cells.

The tail comparison is most pronounced against DDN-AT. Its weakest cells collapse under ℓ_∞ -oriented attacks, with the bottom-10 cells averaging only 15.602%. AttackDRO++ raises these same cells by +15.710 pp, explaining why it achieves a much more balanced whitebox profile than the ℓ_2 specialist, even though DDN-AT remains competitive on several ℓ_2 cells.

Against PGD-AT, the lift is smaller but still positive: +0.726 pp on the bottom-10 cells. PGD-AT’s weakest cells include several bounded ℓ_2 attacks on difficult classes such as *cat* and *deer*; AttackDRO++ reduces many of these failures, indicating that multi-attack training and the Cluster-DRO component help mitigate the cross-norm weakness of PGD-only training.

Overall, the bottom- K analysis supports the aggregate conclusion from Section 6.1: AttackDRO++ gives a broad but modest refinement over Multi-AT, substantially corrects the ℓ_∞ tail weakness of DDN-AT, and reduces the ℓ_2 tail weakness of PGD-AT.

6.3 Graybox Robustness Across Surrogate Models

The whitebox results in Section 6.1 show that AttackDRO++ improves average and held-out robustness over Multi-AT on $n = 20$ paired seeds. We next test whether this gain also holds under cross-model transfer, using the graybox protocol of Section 5.8. In this setting, adversarial examples are generated on one checkpoint and evaluated on an independently trained target checkpoint.

6.3.1 Seed Selection and Whitebox Sanity Check

The graybox panel uses five representative seeds, $\{73, 202, 1337, 7777, 12345\}$, applied to all four methods. This subset was chosen to match the full $n = 20$ paired deltas within 0.022 pp on every reported metric, while also including seed 7777, the worst AttackDRO++ seed in the full panel. Thus, the panel is a representative computational subset rather than a favorable one.

The diagonal entries of the graybox table correspond to the case where the surrogate and target are the same checkpoint. We use these entries as a whitebox sanity check. They need not match the main whitebox evaluation exactly, because the graybox cache uses a balanced 125-image-per-class-per-attack split instead of the full test set for each attack. However, they should preserve the same method ordering and remain close in absolute value.

Table 6.10: Whitebox sanity check from the graybox pipeline. The $n = 5$ columns use diagonal graybox entries, where surrogate and target are the same checkpoint. The $n = 20$ columns report the main whitebox results from Table 6.1.

Target method	Mean(8) $n = 5$	Worst(8) $n = 5$	Mean(8) $n = 20$	Worst(8) $n = 20$
PGD-AT	61.524	50.384	60.499	49.656
DDN-AT	56.922	25.568	56.226	25.957
Multi-AT	62.458	45.584	62.045	45.082
AttackDRO++ (Ours)	62.642	45.088	62.324	45.079

The Mean(8) ordering is identical on both panels — AttackDRO++ > Multi-AT > PGD-AT > DDN-AT — and absolute values agree to within 1.0 pp. More importantly, the paired Mean(8) delta of AttackDRO++ minus Multi-AT is +0.184 pp on the $n = 5$ graybox panel and +0.279 pp on the full $n = 20$ panel: both are positive and of the same order of magnitude. The graybox cache is therefore consistent with the main evaluation protocol and supports the cross-method analyses below.

6.3.2 Cross-Method Transfer Matrix

We aggregate the per-attack per-class transfer table into a method-level 4×4 transfer matrix. Rows index the target method being evaluated, and columns index the surrogate method that generates adversarial examples. Each cell reports the target robust accuracy averaged over all matching target–surrogate seed pairs and the eight evaluation attacks. Higher values indicate that the row target is more robust to adversarial examples crafted on the column surrogate.

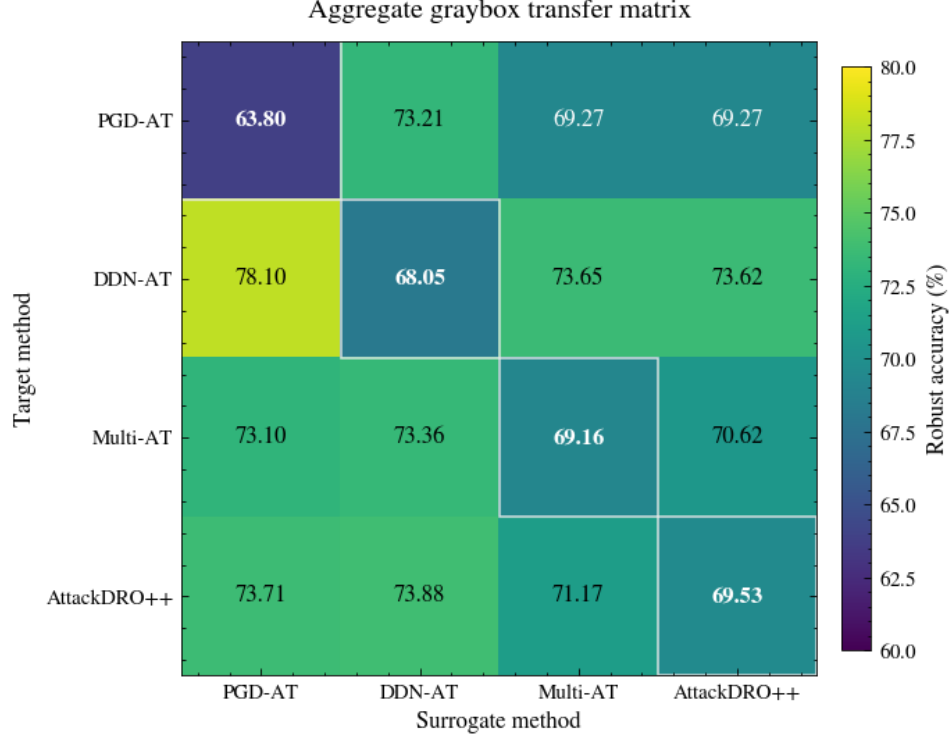


Figure 6.11: Method-level graybox transfer matrix on the $n = 5$ panel. Rows are target methods and columns are surrogate methods. Cell values are robust accuracy (%) averaged over seeds and the eight evaluation attacks.

Two structural patterns are visible. First, rows ranked by mean off-diagonal accuracy give DDN-AT (75.12%) > AttackDRO++ (72.92%) > Uniform Multi-AT (72.36%) > PGD-AT (70.59%). The DDN-AT row is highest not because DDN-AT is the strongest defense — its whitebox Mean(8) is the lowest of the four — but because ℓ_∞ adversarial examples generated by other surrogates transfer poorly to a DDN-AT target whose decision boundary is shaped by ℓ_2 training. Section 6.3.5 quantifies this structural inflation explicitly through the whitebox–graybox gap. Second, the AttackDRO++ row is consistently above the Uniform Multi-AT row for every surrogate, with cross-method gaps ranging from +0.17 pp (DDN-AT surrogate) to +0.61 pp (PGD-AT surrogate). The whitebox advantage of AttackDRO++ over Uniform Multi-AT therefore persists when

adversarial examples are crafted on independently trained surrogates.

The aggregate transfer matrix hides important attack-family structure. Figures 6.12 and 6.13 therefore disaggregate the transfer behavior into the four ℓ_∞ -oriented attacks and the four ℓ_2 -oriented attacks. This grouping makes the specialist behavior more explicit.

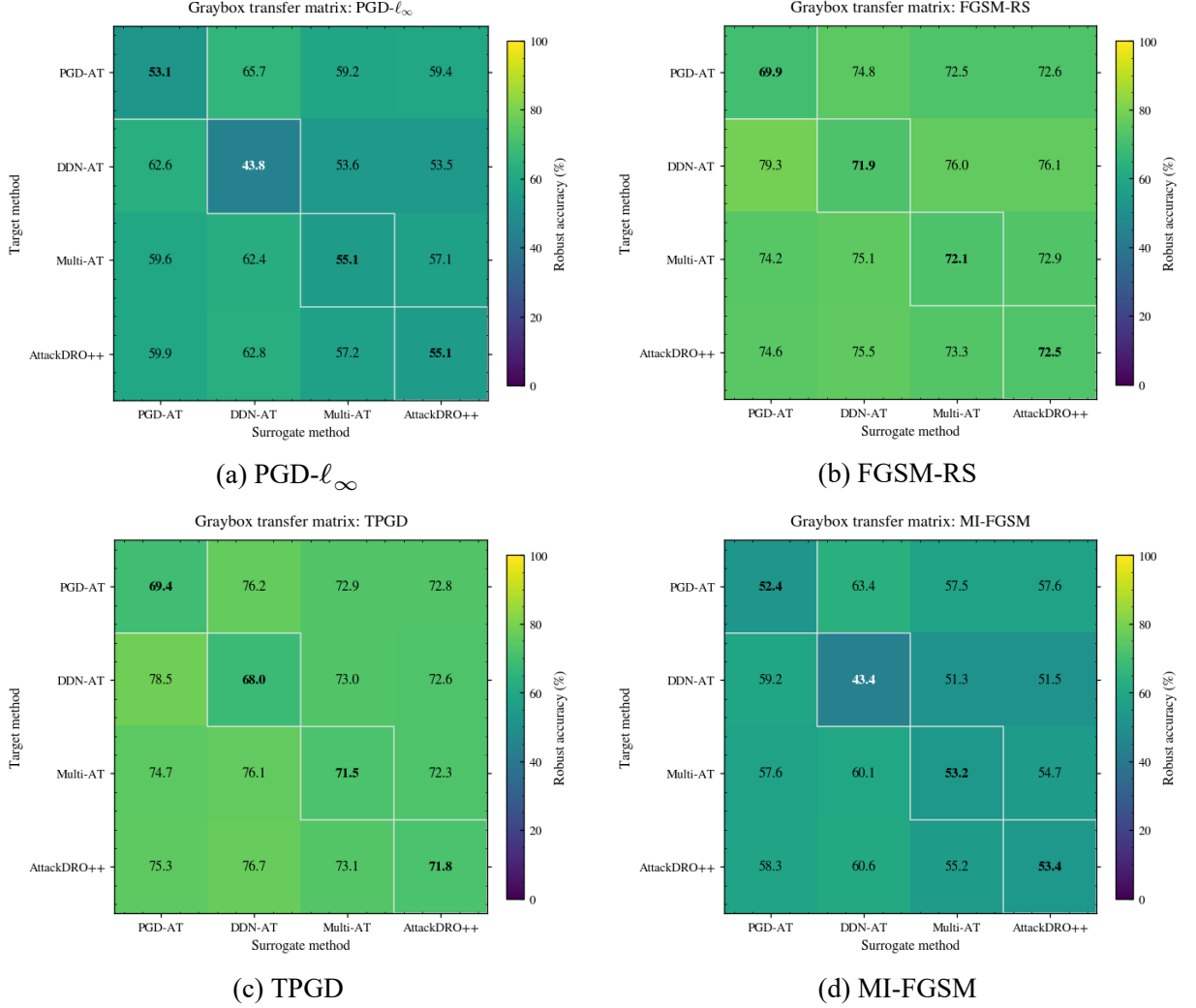


Figure 6.12: Per-attack graybox transfer matrices for the ℓ_∞ -oriented attacks. Rows are target methods and columns are surrogate methods; cells report robust accuracy (%).

On the ℓ_∞ attacks, this advantage disappears: PGD-AT and AttackDRO++ are more competitive, and AttackDRO++ remains closer to the balanced Multi-AT profile. Thus, the per-attack matrices show that DDN-AT's strong graybox row average should be interpreted as family-specific transfer specialization rather than uniform robustness.

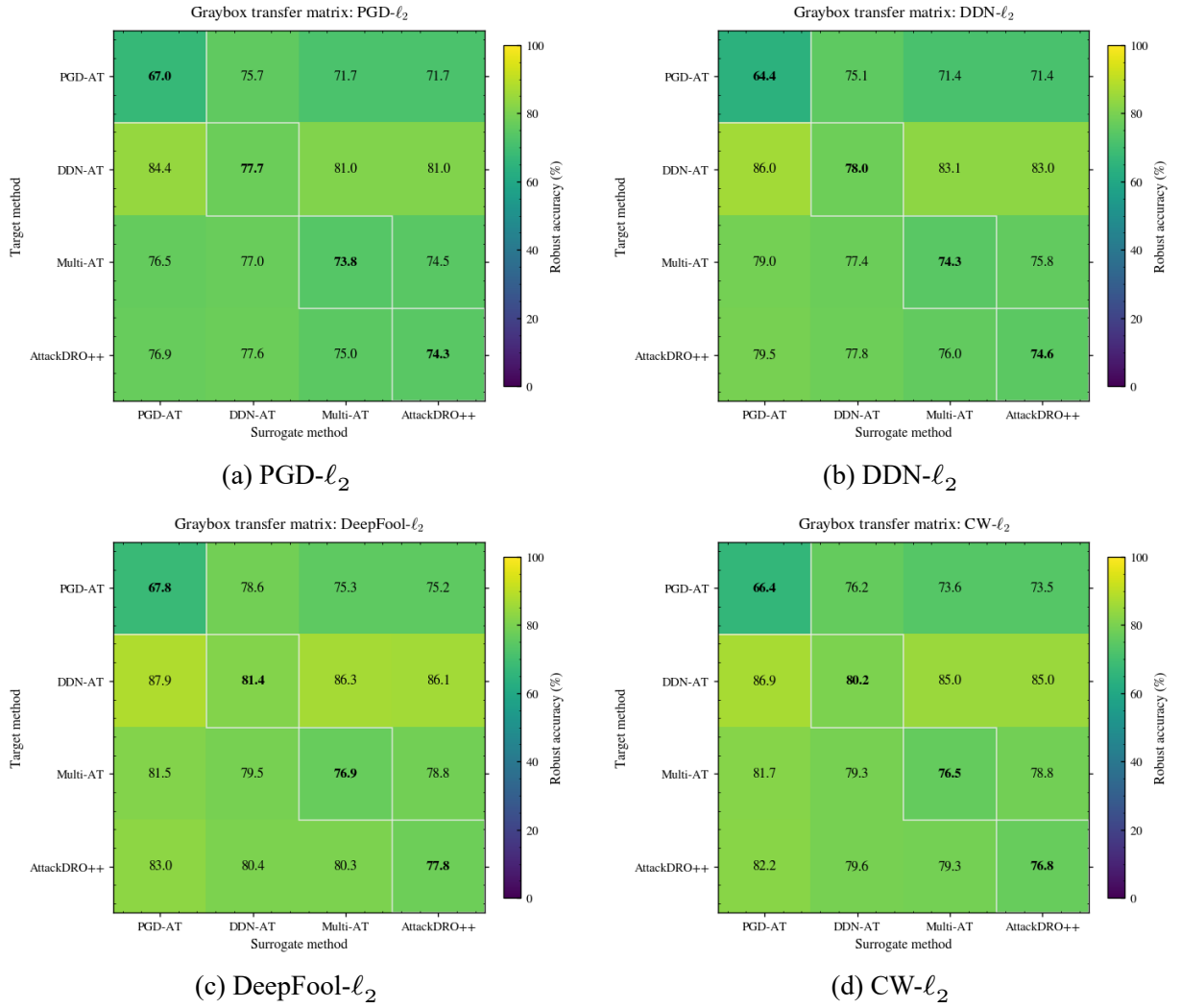


Figure 6.13: Per-attack graybox transfer matrices for the ℓ_2 -oriented attacks. Rows are target methods and columns are surrogate methods; cells report robust accuracy (%).

On the ℓ_2 attacks, the DDN-AT target row is consistently strong, especially when attacks are transferred from PGD-AT, Multi-AT, or AttackDRO++ surrogates. This indicates that DDN-AT’s high aggregate graybox score is driven mainly by ℓ_2 -family transfer behavior, rather than by uniformly strong robustness across both attack geometries. AttackDRO++ remains closer to the balanced Multi-AT profile, while still improving over Multi-AT on most ℓ_2 -oriented transfer cells.

6.3.3 AttackDRO++ vs. Multi-AT

The main graybox comparison is between AttackDRO++ and Multi-AT. Both methods train on the same source attacks $\{\text{PGD-}\ell_\infty, \text{DDN-}\ell_2\}$ and use the same backbone and schedule; they differ only in the use of latent clustering, gradient fingerprints, and the anchored Cluster-DRO objective.

We use the symmetric paired test that has already been described in Section 5.8: for every

triple (target seed, surrogate seed, attack), we pair the configuration “AttackDRO++ as target, Multi-AT as surrogate” against its mirror “Multi-AT as target, AttackDRO++ as surrogate” and compute the per-triple difference. As the results, $n = 5 \times 5 \times 8 = 200$ paired pairs make the comparison invariant to which method is chosen as the surrogate.

Table 6.11: Paired graybox comparisons. Each row uses $n = 200$ paired target–surrogate–attack triples. Positive Δ favors the reference method named in the panel heading.

Compared method	Accuracy		Δ	d_z	t	p
	Reference	Baseline	(pp)			
<i>AttackDRO++ (Ours) vs. baselines</i>						
Multi-AT	71.17	70.62	+0.549	0.42	5.96	1.1×10^{-8}
PGD-AT	73.71	69.27	+4.435	1.33	18.78	6.1×10^{-46}
DDN-AT	73.88	73.62	+0.258	0.04	0.60	0.549
<i>Multi-AT vs. single-attack baselines</i>						
PGD-AT	73.10	69.27	+3.829	1.19	16.89	2.7×10^{-40}
DDN-AT	73.36	73.65	−0.292	−0.05	−0.67	0.504

The AttackDRO++ vs. Multi-AT contrast is the headline result of this section: under symmetric cross-method graybox transfer, AttackDRO++ improves on Multi-AT by +0.549 pp at $p \approx 1.1 \times 10^{-8}$, with a small-to-medium paired effect size of $d_z \approx 0.42$. This effect is roughly twice the corresponding whitebox delta of +0.279 pp on the same seed panel and is significant at far stricter levels. The Cluster-DRO improvement therefore *strengthens* rather than disappears under graybox transfer, indicating that the gain originates from a broader cross-attack robustness profile rather than from a defense narrowly tuned to AttackDRO++’s own gradients.

The single-attack contrasts confirm the expected specialist–versus–balanced pattern. AttackDRO++ improves on PGD-AT by +4.44 pp at $p < 10^{-45}$ ($d_z = 1.33$, a large effect), tracking the whitebox trend that PGD-AT loses heavily on the ℓ_2 family it does not train against. The AttackDRO++ vs. DDN-AT contrast is statistically indistinguishable from zero (+0.26 pp, $p = 0.55$); this reflects the DDN-AT graybox inflation discussed above, not actual parity. We dissect this asymmetry in Section 6.3.5.

6.3.4 Per-Attack Analysis

To check whether the +0.549 pp Mean(8) gain over Multi-AT is broad or concentrated, we report paired graybox deltas attack by attack on the same $n = 200$ pair set. Each attack contributes 25 paired triples (one per surrogate–target seed combination).

Table 6.12: Per-attack symmetric graybox comparison between AttackDRO++ and Multi-AT. Each attack uses $n = 25$ paired seed combinations. Bold deltas are significant at $p < 0.05$.

Attack	Family	A mean (%)	B mean (%)	Δ (pp)	p
FGSM-RS	held-out ℓ_∞	73.32	72.86	+0.454	0.0114
PGD- ℓ_∞	seen ℓ_∞	57.17	57.07	+0.102	0.307
TPGD	held-out ℓ_∞	73.09	72.30	+0.790	0.0240
MI-FGSM	held-out ℓ_∞	55.20	54.75	+0.458	0.0022
PGD- ℓ_2	held-out ℓ_2	74.97	74.53	+0.432	0.0373
DDN- ℓ_2	seen ℓ_2	76.04	75.79	+0.256	0.341
DeepFool- ℓ_2	held-out ℓ_2	80.29	78.82	+1.469	0.0002
CW- ℓ_2	held-out ℓ_2	79.26	78.83	+0.429	0.249

AttackDRO++ produces a positive paired delta on all eight attacks. Five of these deltas are individually significant at $p < 0.05$, and the gain is largest on DeepFool- ℓ_2 (+1.47 pp, $p = 1.6 \times 10^{-4}$), the most local-gradient ℓ_2 attack and the held-out attack on which the whitebox gain over Multi-AT was also largest. The two seen attacks (PGD- ℓ_∞ and DDN- ℓ_2) contribute small, non-significant gains, consistent with the fact that both methods already train against these attacks. The graybox Mean(8) gain is therefore distributed across attack families rather than concentrated on one family.

We aggregate the per-attack deltas into the four attack groups defined in Section 5.7 to compare the cross-method graybox profile across all four methods.

Table 6.13: Cross-method graybox accuracy by attack family. Values are off-diagonal target accuracy (%) averaged over the $n = 5$ panel. Bold values mark the best method in each column.

Target method	Seen ℓ_∞	Seen ℓ_2	Heldout ℓ_∞	Heldout ℓ_2	Mean(8)
PGD-AT	61.48	72.61	68.93	74.60	70.59
DDN-AT	56.58	84.03	68.60	84.85	75.12
Multi-AT	59.71	77.40	68.64	78.62	72.36
AttackDRO++ (Ours)	59.96	77.78	69.18	79.35	72.92

The family decomposition makes three points concrete. The held-out ℓ_∞ column has AttackDRO++ at the top (69.18%) by a small but consistent margin over the other three methods. The held-out ℓ_2 column is dominated by DDN-AT (84.85%), with AttackDRO++ (79.35%) ranked second and improving on Multi-AT by +0.73 pp. The seen ℓ_2 column shows the same pattern. The seen ℓ_∞ column favors PGD-AT, again by structure: surrogates generate strong PGD- ℓ_∞ adversarial examples, and only PGD-AT is robust to them. AttackDRO++ is the only method that is at or above Multi-AT on every attack family, while DDN-AT and PGD-AT each remain narrow specialists under transfer.

6.3.5 Whitebox–Graybox Gap and Specialist Inflation

The DDN-AT target row in Figure 6.11 and the DDN-AT row in Table 6.13 are the highest under graybox transfer, even though DDN-AT has the lowest whitebox Mean(8) on both the $n = 5$ and the full $n = 20$ panels. We resolve this apparent contradiction by quantifying the gap between whitebox and cross-method graybox accuracy for each method. A large positive gap indicates that a defense looks much stronger under graybox transfer than under direct whitebox attack, which is a signature of a structural transfer mismatch rather than a broader robustness profile.

Table 6.14: Whitebox–graybox gap by target method on the $n = 5$ panel. The gap is cross-method graybox Mean(8) minus diagonal whitebox Mean(8).

Target method	Whitebox Mean(8)	Graybox Mean(8)	Gap (pp)
PGD-AT	61.52	70.59	+ 9.06
DDN-AT	56.92	75.12	+ 18.20
Multi-AT	62.46	72.36	+ 9.90
AttackDRO++ (Ours)	62.64	72.92	+10.27

PGD-AT, Multi-AT, and AttackDRO++ all show a comparable whitebox–graybox gap of approximately 9–10 pp, indicating that adversarial examples crafted on alternate surrogates are weaker than those crafted directly on the target by a roughly constant amount. DDN-AT, by contrast, gains +18.20 pp under graybox transfer, almost twice the gap of any other method. Its row-leading graybox accuracy is therefore not evidence of a stronger defense; it is an artifact of how poorly ℓ_∞ adversarial examples generated by PGD-AT, Multi-AT, and AttackDRO++ surrogates transfer to a purely- ℓ_2 -trained target. AttackDRO++ retains the highest whitebox Mean(8) of the four methods while exhibiting a graybox gap that is only +0.37 pp larger than that of Multi-AT — consistent with a balanced cross-attack defense rather than a transfer-favored specialist.

The per-(class $\times$ attack) whitebox–graybox gap heatmaps in Figures 6.14–6.17 make the specialist-inflation effect concrete. Cell values are computed as graybox accuracy minus whitebox accuracy, so positive values indicate that transferred attacks are weaker than direct whitebox attacks on the same target.

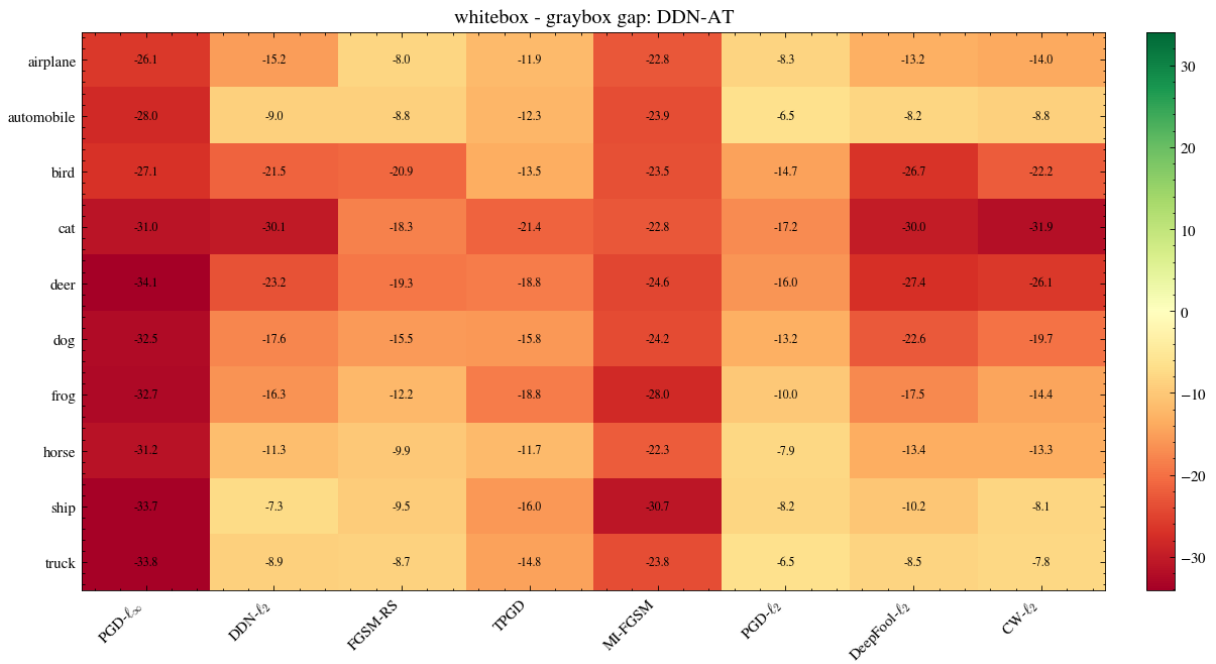


Figure 6.14: Whitebox–graybox gap heatmap for DDN-AT.

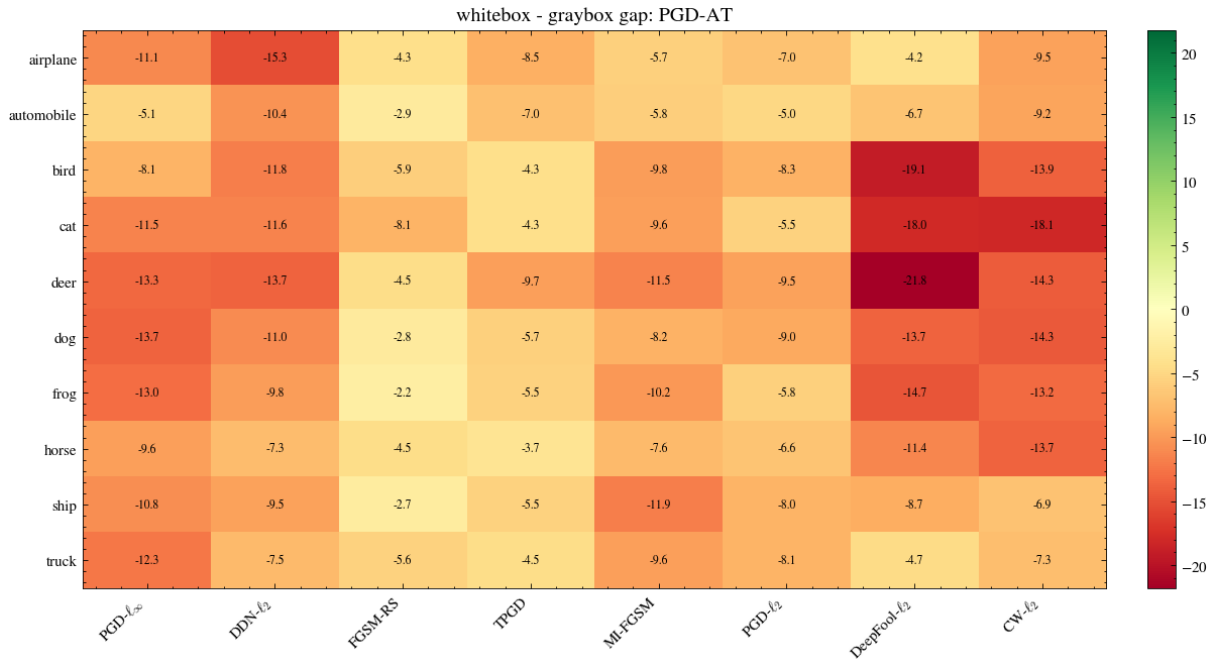


Figure 6.15: Whitebox–graybox gap heatmap for PGD-AT.

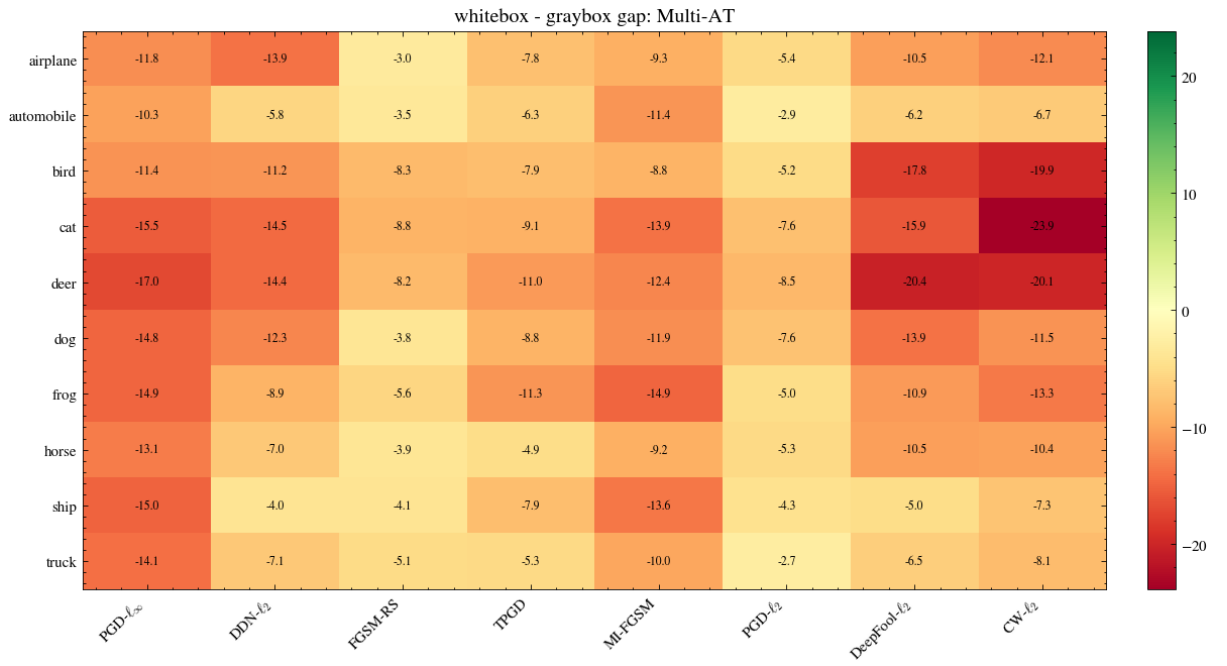


Figure 6.16: Whitebox–graybox gap heatmap for Multi-AT.

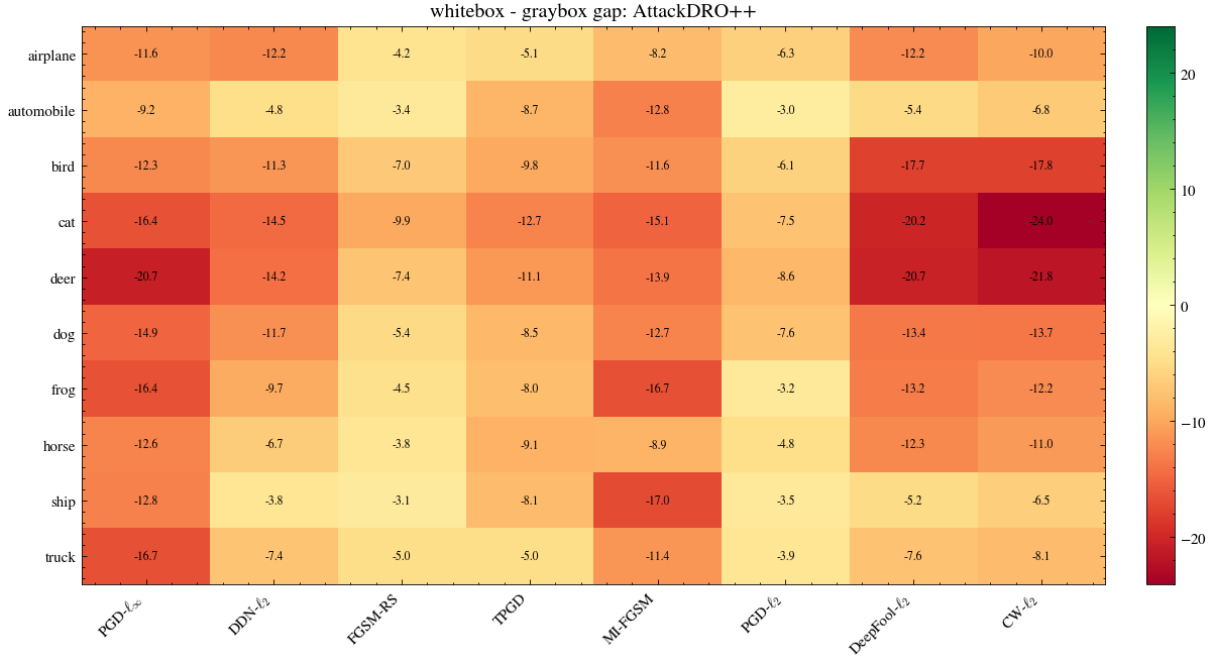


Figure 6.17: Whitebox–graybox gap heatmap for AttackDRO++ (Ours).

DDN-AT exhibits broad positive gaps on the ℓ_∞ -oriented attack columns, indicating that its high graybox score is partly driven by weak transfer from ℓ_∞ surrogates rather than uniformly strong robustness. PGD-AT, Multi-AT, and AttackDRO++ show smaller and more balanced gap patterns, with AttackDRO++ avoiding the severe specialist inflation observed for DDN-AT.

6.3.6 Per-(Class-Attack) Analysis

We complement the aggregate analysis with a paired per-cell test. For each of the 80 attack–class cells, we compare the graybox accuracy of the AttackDRO++ target against each baseline target across the five training seeds (each cell pairs $n = 5$ seeds). A two-sided paired t -test is computed per cell.

Table 6.15: Per-(class $\times$ attack) graybox tests comparing AttackDRO++ against each baseline. Δ is computed as AttackDRO++ minus the baseline.

Baseline	Positive cells ($\Delta > 0$)	Sign-test p	Mean Δ (pp)	Significant cells AttackDRO++ / Baseline	Bottom-10 lift (pp)
Multi-AT	59/80	1.3×10^{-5}	+0.557	2 / 3	+0.779
PGD-AT	52/80	4.8×10^{-3}	+2.332	34 / 12	+1.329
DDN-AT	27/80	2.4×10^{-3}	−2.206	16 / 32	+0.709

Three observations follow. First, AttackDRO++ exceeds Multi-AT on 59 of 80 cells with a mean per-cell delta of +0.557 pp, in close agreement with the +0.549 pp aggregate from Table 6.11. Cell-level significance is sparse (2 vs. 3 cells reach $p < 0.05$ in either direction), as expected under the $n = 5$ per-cell sample size and the small absolute deltas; the bulk of the evidence for AttackDRO++ is therefore the *distributional* shift across many cells rather than a few cell-level landslides.

Second, the bottom-10 lift is positive in all three contrasts. The cells on which Multi-AT, PGD-AT, and DDN-AT are weakest concentrate on the cat, deer, dog, and bird classes under MI-FGSM, PGD- ℓ_∞ , and PGD- ℓ_2 . AttackDRO++ improves on Multi-AT by +0.779 pp on Multi-AT’s worst 10 cells, and improves on PGD-AT by +1.329 pp on PGD-AT’s worst 10 cells.

Third, the AttackDRO++ vs. DDN-AT cell-level comparison (−2.21 pp on average, 32 cells favoring DDN-AT at $p < 0.05$) is the per-cell counterpart of the DDN-AT graybox inflation: most of DDN-AT’s wins are on ℓ_2 cells, while AttackDRO++ retains the wins it needs on the ℓ_∞ cells that DDN-AT collapses on under whitebox attack (Section 6.1.2).

The heatmaps in Figures 6.18–6.20 visualize these patterns explicitly: green cells favor AttackDRO++, red cells favor the baseline. The AttackDRO++ vs. Multi-AT heatmap is broadly green and small in magnitude; the AttackDRO++ vs. PGD-AT heatmap is broadly green with a few large red cells on ℓ_∞ attacks where PGD-AT specializes; and the AttackDRO++ vs. DDN-AT heatmap is mixed.

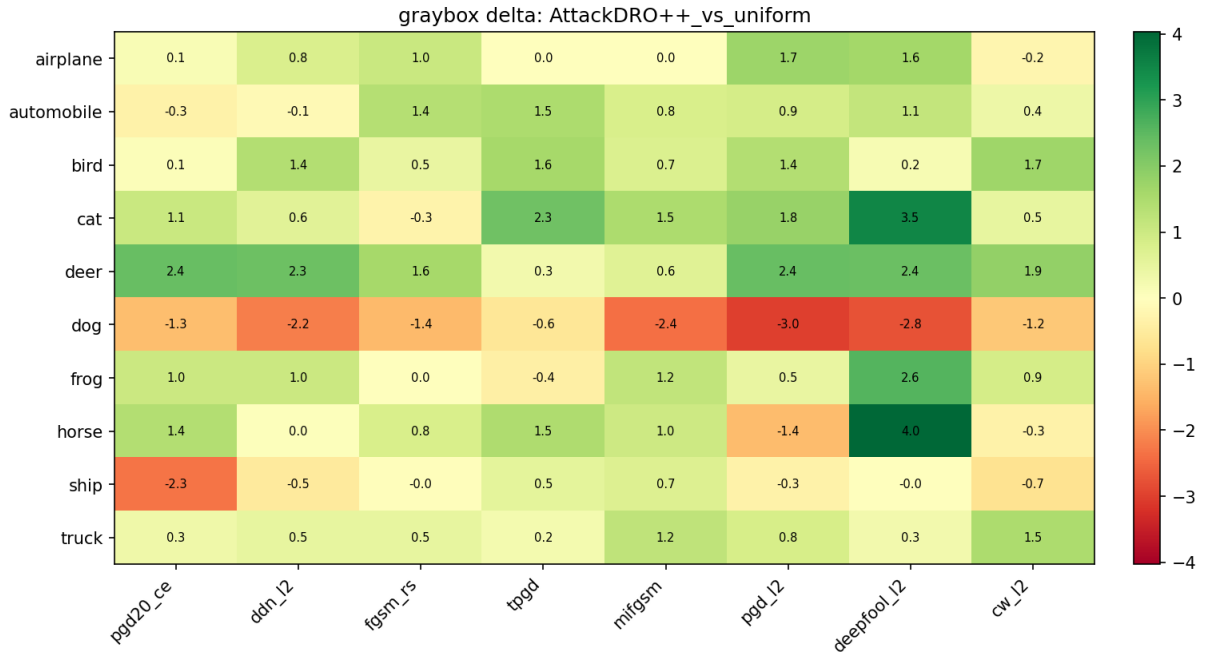


Figure 6.18: Per-(class $\times$ attack) graybox delta heatmap for AttackDRO++ (Ours) minus Multi-AT. Green favors AttackDRO++ (Ours); red favors Multi-AT.

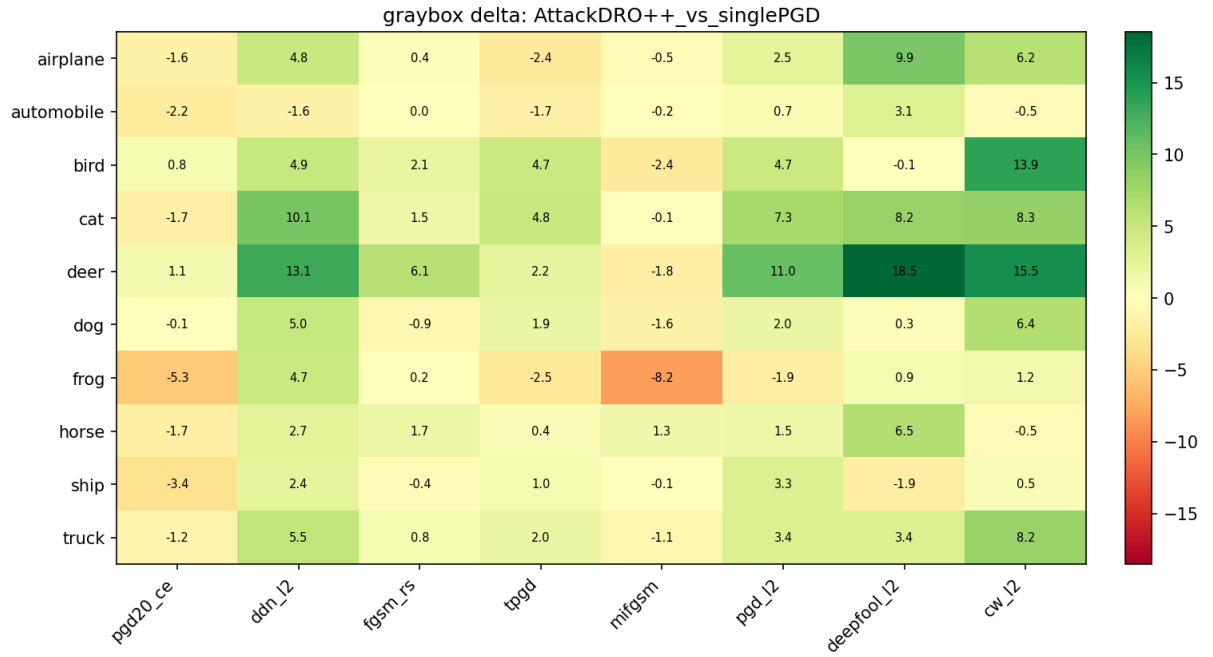


Figure 6.19: Per-(class $\times$ attack) graybox delta heatmap for AttackDRO++ (Ours) minus PGD-AT. Green favors AttackDRO++ (Ours); red favors PGD-AT.

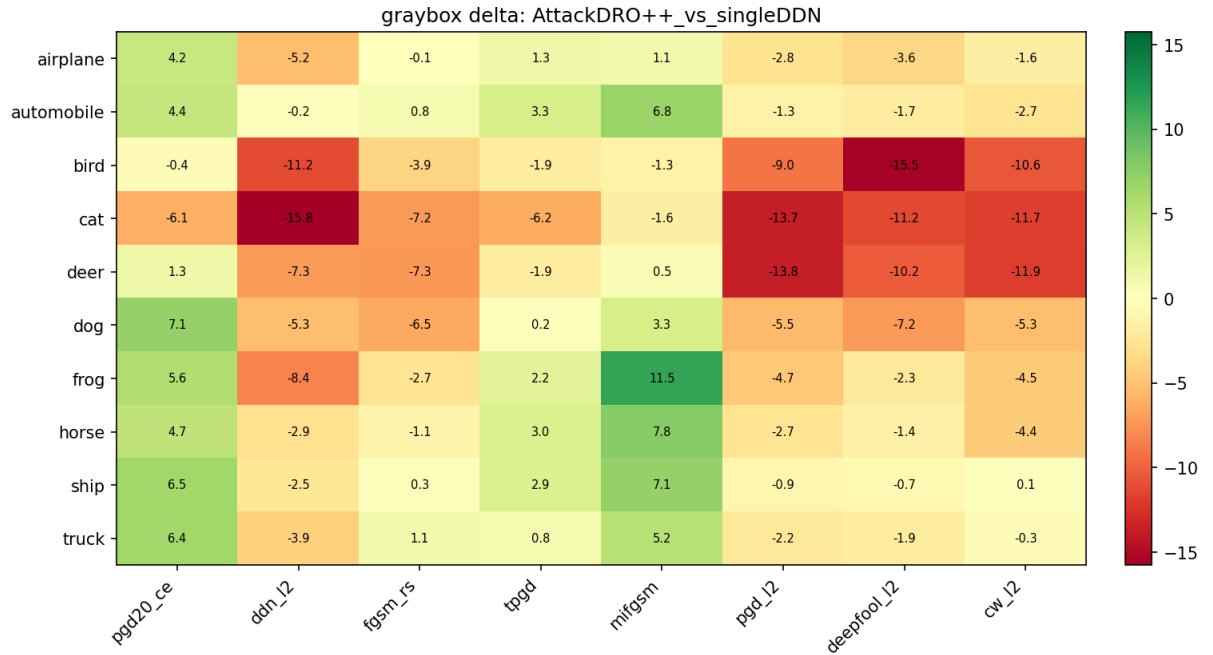


Figure 6.20: Per-(class $\times$ attack) graybox delta heatmap for AttackDRO++ (Ours) minus DDN-AT. Green favors AttackDRO++ (Ours); red favors DDN-AT.

6.3.7 Method Contribution Decomposition

The whitebox results of Section 6.1 compare AttackDRO++ against Multi-AT and against the two single-attack baselines as three separate contrasts. Under graybox transfer, we can additionally ask how much of the AttackDRO++ improvement over each single-attack baseline is contributed by multi-attack training (the Multi-AT component) versus the Cluster-DRO objective (the AttackDRO++ component on top of Multi-AT). We decompose the total AttackDRO++ vs. single-attack delta as

$$\Delta_{\text{total}} = \underbrace{(\text{Multi-AT}) - (\text{single-AT})}_{\Delta_{\text{multi-attack}}} + \underbrace{(\text{AttackDRO++}) - (\text{Multi-AT})}_{\Delta_{\text{Cluster-DRO}}},$$

and report this decomposition for whitebox Mean(8), cross-method graybox Mean(8), and the bottom-10-cell cross-method graybox average (the average graybox accuracy on the ten attack–class cells in which the target is weakest). The bottom-10 metric is included because it isolates worst-cell behavior, which is most sensitive to the AttackDRO++ objective.

Table 6.16: AttackDRO++ (Ours) gain decomposition. Values are percentage-point changes.

Metric	Base	Multi-AT	DRO++	Total
Whitebox Mean(8)	PGD-AT	+0.934	+0.184	+1.118
Whitebox Mean(8)	DDN-AT	+5.536	+0.184	+5.720
Cross-graybox Mean(8)	PGD-AT	+1.775	+0.557	+2.332
Cross-graybox Mean(8)	DDN-AT	−2.763	+0.557	−2.206
Bottom-10 graybox	PGD-AT	+0.530	+0.799	+1.329
Bottom-10 graybox	DDN-AT	−0.090	+0.799	+0.709

The Cluster-DRO effect (column 4) is positive on every metric and decomposition path, ranging from +0.184 pp on whitebox Mean(8) to +0.799 pp on the bottom-10 cross-graybox metric. Two implications follow. First, the Cluster-DRO effect is larger under graybox transfer than under direct whitebox attack, consistent with the +0.279 pp → +0.549 pp scaling observed in Section 6.3.3. Second, the Cluster-DRO effect is roughly four times larger on the bottom-10 cells (+0.799 pp) than on whitebox Mean(8), supporting the design intent that the Cluster-DRO objective primarily reshapes worst-cell behavior rather than the average.

The multi-attack column tells a complementary specialist story. Going from single-AT to Multi-AT adds +5.54 pp of whitebox Mean(8) when the baseline is DDN-AT, but removes -2.76 pp of cross-graybox Mean(8) for the same baseline — because DDN-AT’s graybox accuracy is inflated by the very ℓ_∞ transfer mismatch that Multi-AT corrects. The cluster-DRO component remains positive in both cases, indicating that it adds robustness that is not absorbed by either multi-attack training alone or DDN-AT’s structural graybox advantage.

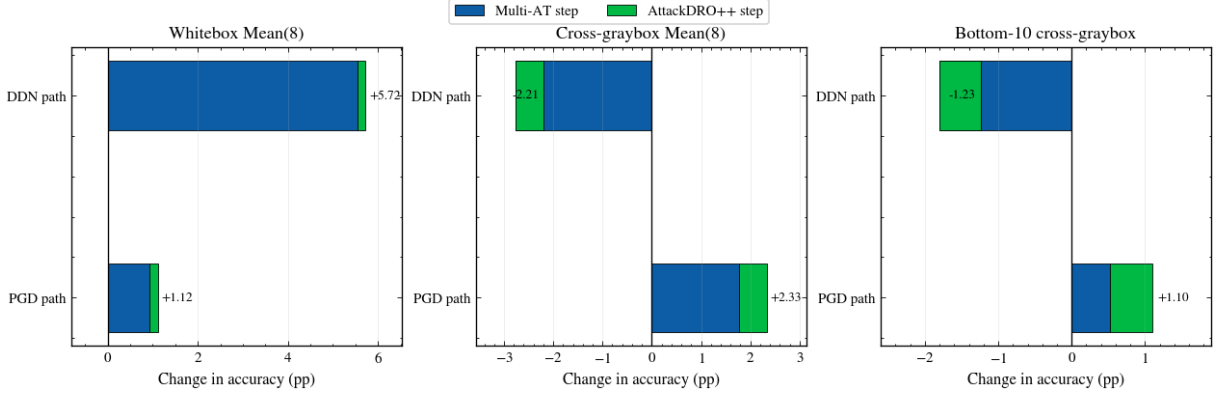


Figure 6.21: Visual decomposition of AttackDRO++ improvement over the single-attack baselines into a multi-attack effect (Multi-AT minus single-AT) and a Cluster-DRO effect (AttackDRO++ minus Multi-AT) across whitebox, cross-method graybox, and bottom-10 cross-method graybox metrics, for the two single-AT decomposition paths.

6.4 Sensitivity to Key Hyperparameters

6.4.1 Anchor Strength

The anchor-strength ablation examines the interpolation between the uniform multi-attack ERM loss and the adaptive Cluster-DRO loss. A smaller anchor gives more weight to the uniform baseline, while a larger anchor increases the influence of cluster-level reweighting. Figure 6.22 provides the dashboard view of the ablation results, with the anchor-strength comparison shown as the first sensitivity check. Figure 6.22 reports the corresponding numeric summary.

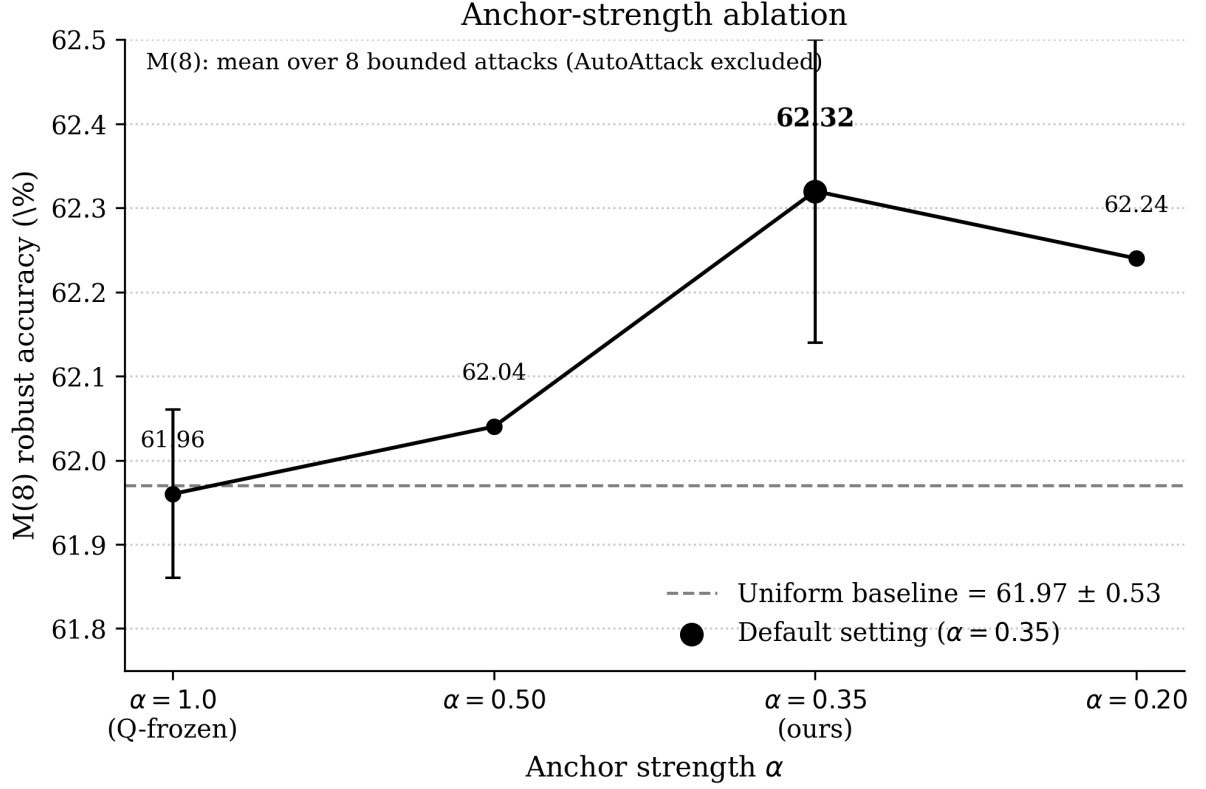


Figure 6.22: Ablation dashboard for AttackDRO++. The dashboard summarizes the sensitivity of M(8) robust accuracy to anchor strength, number of clusters, and cluster-refresh schedule. M(8) denotes the mean robust accuracy over the eight bounded evaluation attacks and excludes AutoAttack.

Overall, the ablation supports using a moderate anchor rather than relying entirely on either uniform ERM or aggressive Cluster-DRO weighting. The default $\lambda_{\text{DRO}} = 0.35$ is retained because it provides a practical balance: the uniform multi-attack signal remains dominant, while the cluster-level correction can still emphasize harder latent groups. Since some settings are evaluated with fewer seeds, the observed differences should be interpreted as a sensitivity trend rather than a definitive ranking.

6.4.2 Number of Clusters

The number of clusters controls the granularity of the latent groups used by AttackDRO++. Too few clusters may merge distinct failure modes, while too many clusters may fragment the minibatch and make group-loss estimates noisy. Table 6.17 summarizes the sensitivity to this design choice.

Table 6.17: Number-of-clusters ablation for AttackDRO++. The number of clusters K determines the granularity of the latent adversarial groups used by Cluster-DRO. $M(8)$ is the mean robust accuracy over the eight bounded evaluation attacks and excludes AutoAttack.

Setting	K	$M(8)$	Notes
Coarse grouping	2	TODO	TODO
Default grouping	4	TODO	Default setting
Finer grouping	6	TODO	TODO
Very fine grouping	10	TODO	TODO

The default choice $K = 4$ is kept as a conservative middle point. It is expressive enough to separate more than the two source attack identities, but it avoids the instability that can arise when clusters become too small. The results should be read as evidence that the method is reasonably robust around the default cluster granularity, not as proof that $K = 4$ is universally optimal.

6.4.3 Cluster-Refresh Schedule

We studies the effect of the cluster-refresh interval R in AttackDRO++ with Anchor35, GradFP, and $K = 4$. The refresh interval controls how often the K-means clusterer is re-fitted from the feature bank. A shorter interval may adapt more quickly to representation changes, while a longer interval may provide a more stable clustering signal.

Table 6.18 compares refreshing the clusterer every one epoch and every two epochs. The two settings produce very similar results. The default two-epoch schedule gives a slightly higher Mean(8), with 62.44 ± 0.22 , compared with 62.38 ± 0.14 for the one-epoch schedule. It also achieves a slightly higher ℓ_∞ -family average. In contrast, refreshing every epoch gives slightly better Worst(8), AutoAttack, and ℓ_2 -family average.

Overall, the differences are small relative to the three-seed setting. Therefore, this ablation does not show a strong advantage for either refresh interval. We retain $R = 2$ epochs as the default because it gives comparable robustness while avoiding overly frequent cluster refitting.

Table 6.18: Cluster-refresh schedule ablation for AttackDRO++ with Anchor35, GradFP, and $K = 4$. Values are robust accuracy (%) reported as mean $\pm$ standard deviation over three seeds. Mean(8) averages over the eight bounded evaluation attacks and excludes AutoAttack. Worst(8) is the minimum robust accuracy over the same eight bounded attacks and also excludes AutoAttack. AutoAttack is reported separately.

Refresh interval	Seeds	Mean(8) $\uparrow$	Worst(8) $\uparrow$	AutoAttack $\uparrow$	ℓ_2 -family avg $\uparrow$	ℓ_∞ -family avg $\uparrow$
1 epoch	3	62.38 $\pm$ 0.14	45.17 $\pm$ 0.34	40.30 $\pm$ 1.37	67.57 $\pm$ 0.16	57.18 $\pm$ 0.27
2 epochs	3	62.44 $\pm$ 0.22	45.14 $\pm$ 0.23	39.45 $\pm$ 0.68	67.56 $\pm$ 0.08	57.31 $\pm$ 0.35

Chapter 7

Conclusion and Future Work

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Appendix A

Illustrative Pseudo-code and Flowcharts

Appendix B

Experimental Logs and Configurations

Appendix C

Additional Tables and Visualizations

Appendix D

Compute Resources and Environment Notes